Economic and Political Uncertainty Index for Peru Using X and DeepSeek-V3

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Abstract

We propose the first daily index of economic and political uncertainty (EPU) for Peru, and its defining novelty is the source: rather than newspapers, we read the real-time discourse on X (formerly Twitter). Dominant policy-uncertainty indices count newspaper articles and are typically monthly and lagged; X instead records, at daily frequency, how the country's most influential voices in politics, economics, journalism, and business react to events as they happen. We classify their messages along economic, political, and uncertainty dimensions with a large language model (LLM), DeepSeek-V3, and recombine these dimensions into a family of complementary daily indices, among them a broader political-uncertainty (PU) index. We then ask how closely the index behaves like two established market-based measures of uncertainty, the Chicago Board Options Exchange Volatility Index (VIX) and the volatility of the Bolsa de Valores de Lima (BVL). It co-moves with both at a moderate level, and more strongly within documented crises. This is what one expects of measures that share a common uncertainty component but differ in what else they capture: the index reads domestic political and economic uncertainty, the VIX global financial risk, and BVL volatility the Lima equity market. The moderate co-movement is therefore not a shortcoming but evidence that the index is not redundant with the benchmarks.

Keywords: Economic Political Uncertainty, X (formerly Twitter), Large Language Models, social media, DeepSeek-V3, GARCH, Peru.

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1 Introduction

Uncertainty about economic and political conditions shapes real and financial outcomes through well-documented channels (Bloom, 2009, 2014). When uncertainty rises, firms postpone irreversible investment and hiring while investors reprice risk, so uncertainty shocks act much like adverse demand shocks for output and employment. These effects are amplified in emerging economies, where external financing is procyclical and sovereign-risk premia react sharply to political news (Frankel, 2010; Carrière-Swallow and Céspedes, 2013). Measuring uncertainty at high frequency is therefore valuable to policymakers and investors alike.

Much of the relevant information arrives as narrative, and how it propagates through public discourse is itself an economic force: narratives spread contagiously and drive aggregate fluctuations (Shiller, 2017), and media coverage can generate frenzies and comovement in asset prices (Veldkamp, 2006). Empirically, the tone of the financial press predicts short-run stock returns, especially in downturns (Tetlock, 2007; García, 2013), and high-frequency proxies built from internet search (Da et al., 2015) and social-media text (Baker et al., 2021; Tumasjan et al., 2010; Bovet and Makse, 2019) track sentiment and forecast returns and volatility.

The dominant measure of policy uncertainty is the newspaper-based index, as in Baker et al. (2016), which counts articles jointly mentioning the economy, policy, and uncertainty; it relies on traditional media and is typically available only monthly. To capture real-time dynamics, recent work leverages social-media platforms such as X (formerly Twitter); for instance, Becerra and Sagner (2023) use the official X handles of mainstream Chilean news outlets, newspapers, and radio stations. While this provides a higher-frequency alternative, it remains bound to centralized, institutional media narratives. We depart from this institutional focus by tracking the accounts of key individual opinion leaders (politicians, journalists, analysts, and policy influencers) capturing a more direct and responsive measure of uncertainty as it unfolds.

Turning such discourse into a quantitative index is a measurement problem that the text-as-data literature has progressively addressed (Gentzkow et al., 2019). Early measures relied on fixed dictionaries and word counts (Loughran and McDonald, 2011; Baker et al., 2016; Becerra and Sagner, 2023), which are transparent but miss context, negation, and irony; supervised and contextual models then improved accuracy (Hassan et al., 2019). The transformer architecture behind modern large language models (from Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2019) to the generative models that followed (Brown et al., 2020)) now classifies short, informal text with high reliability in benchmark studies, opening new possibilities for economic measurement (Korinek, 2023). We exploit this advance: instead of keyword matching, a large language model reads each tweet and judges whether it concerns the economy, politics, and uncertainty.

Our sample runs from January 2018 to January 2023. The setting is informative: within five years Peru had six presidents, a dissolved Congress, a contested election, an attempted self-coup, and sustained protests, all overlapping with the COVID-19 pandemic and the 2021–2022 inflation surge—a concentration of political and economic crises in a single small open economy that makes it a natural laboratory for a near-real-time measure of uncertainty. We do not extend the window in either direction, for reasons that are practical rather than conceptual: classifying the full corpus with a large language model carries a monetary cost that scales with the number of tweets; the platform's data-access terms tightened sharply during 2023, raising the cost of retrieving tweets beyond our window; and before 2018 the platform was used only sparsely by the Peruvian political and economic elite we track, so earlier coverage would be thin and unrepresentative. Within these bounds the index still delivers what no existing measure for Peru does: a daily-frequency reading of economic and political uncertainty, built with a new large-language-model methodology that reads each message in context rather than matching keywords.

This paper makes three contributions. First, we build a new daily index of economic and political uncertainty for Peru from a novel data source: the X discourse of influential accounts in politics, economics, journalism, and business. Unlike the newspaper-based economic policy uncertainty index of Baker et al. (2016), which is monthly and relies on press archives, our index exploits high-frequency social-media text, classifying each tweet along economic, political, and uncertainty dimensions with a large language model (DeepSeek-V3). By economic and political uncertainty (EPU) we mean the joint intersection $E \cap P \cap U$ of economic, political, and uncertainty-laden discourse—a social-media analogue of, but conceptually distinct from, Baker et al. (2016). Recombining these dimensions yields a family of daily indices spanning economic and political uncertainty that does not yet exist for Peru. The construction is portable: it requires only a list of influential accounts and access to the platform's text, so the same pipeline can be replicated for any country in which X hosts an active political and economic conversation.

Second, we ask how similar the index is to existing measures of uncertainty: the Chicago Board Options Exchange Volatility Index (VIX) and the Bolsa de Valores de Lima (BVL). We compare the co-movement of their levels through static (Pearson) and rolling correlation. We also extract the conditional volatility of the BVL return from a generalized autoregressive conditional heteroskedasticity (GARCH) model of order $(1,1)$ (Bollerslev, 1986), and compare it with the index.

Third, we ask whether this similarity is constant or state-dependent. Splitting the sample into calm periods and documented political and economic crisis windows (dated from event chronologies in the tradition of Romer and Romer (2010) and event-based risk indices (Caldara and Iacoviello, 2022)) lets us assess whether the index tracks market-based uncertainty more closely when uncertainty is acute.

The paper is organized as follows. Section 2 describes the data; Section 3 constructs the index family; and Section 4 sets out the methodology. Section 5 defines a set of time periods that correspond to crises in Peru, whether political or economic in nature, including, for example, the pandemic. Section 6 reports the empirical results: the static and rolling correlation of the index with the VIX and BVL volatility, together with the crisis–calm split (Section 6.1), and the conditional volatility of the Lima market obtained from a GARCH(1,1) model, compared as a third series on the same footing with the index and the VIX (Section 6.2). Section 7 concludes.

2 Data

2.1 Tweet Collection

We download tweets from 97 accounts representing the main agents in Peruvian political and economic discourse, spanning 23 January 2018 to 31 January 2023, across five categories: journalists (24), politicians (42), economists (16), political analysts (13), and private-sector representatives (2). Accounts are selected for public visibility (follower count, media citations, and a media presence that is common knowledge in Peruvian society), sustained activity across the window, and coverage of the main institutional actors (Congress, the executive, the central bank, major media outlets, among others). The full set is listed in Table 1. This elite focus limits representativeness relative to a full-population sample but follows Baker et al. (2016) in targeting "informed agents," whose discourse is more likely to reflect genuine policy uncertainty than noise.

The data are obtained via the X (formerly Twitter) application programming interface (API), which yields one comma-separated (CSV) file per account. Figure 1 summarizes the full processing pipeline in four stages. First, in the ingestion stage, all available tweets of each account within the sample window are pulled from the API into per-account raw files—the 101 accounts of the initial download. Second, an Extract–Transform–Load step merges, deduplicates, and cleans these files; it is here that the four institutional accounts (corporate and academic broadcast feeds) are dropped, reducing the 101 ingested accounts to the 97 individual accounts retained for analysis and leaving a consolidated corpus of 213 249 tweets. Third, every surviving tweet is sent to the DeepSeek-V3 API and labelled along the economic, political, and uncertainty dimensions, yielding a tweet-level uncertainty dataset. Fourth, this labelled dataset feeds the econometric analysis. We keep all items—original tweets, retweets, replies, and quoted posts—so that each retained item is a single tweet-level observation; the index therefore measures the intensity of discourse.

2.2 Semantic Classification via a Large Language Model (LLM)

The original Baker et al. (2016) approach classifies a newspaper article as policy-uncertainty-related if it contains at least one keyword from each of three pre-specified dictionaries: an "economic" list D_E (e.g., "economy", "inflation"), a "policy" list D_P (e.g., "Congress", "legislation"), and an "uncertainty" list D_U (e.g., "uncertain", "risk").

Formally, let x_a denote the raw text of article a (a string of tokens) and let $\text{tokens}(x_a)$ be the multi-set of words it contains. Let $w_E \in D_E$ denote any word belonging to the economic dictionary, and define $w_P \in D_P$, $w_U \in D_U$ analogously. Baker et al. (2016) classifier assigns a binary label $z_a^{\text{BBD}} \in \{0, 1\}$ to article a:

$$z_a^{\text{BBD}} = \mathbf{1}\{\exists\, w_E \in D_E,\, w_P \in D_P,\, w_U \in D_U : w_E, w_P, w_U \in \text{tokens}(x_a)\},$$

so $z_a^{\text{BBD}} = 1$ when the article contains at least one representative term of each of the three dictionaries simultaneously. This deterministic rule misses synonyms, idiomatic expressions, irony, and context-dependent meaning.

We replace this rule with a semantic classifier based on DeepSeek-V3. Let x_{ijt} denote the text of the j-th tweet posted by user i on day t, where $i = 1, \ldots, 97$ indexes the X accounts, t indexes calendar days in the sample window, and $j = 1, \ldots, n_{it}$ indexes that user's tweets on day t. The classifier returns a triple

$$z_{ijt} = (E_{ijt},\, P_{ijt},\, U_{ijt}) \in \{0, 1\}^3,$$

where $E_{ijt} = 1$ indicates that tweet (i, j, t) has economic content, $P_{ijt} = 1$ indicates political content, and $U_{ijt} = 1$ indicates the expression of uncertainty.

The three dimensions are produced independently, by three separate expert-level prompts evaluated on the same tweet, each returning a single binary token. Figure 2 shows the anatomy of one such prompt. It is assembled from six blocks: a role description that casts the model as a panel of domain experts (economist, financial analyst, and so on); a task description stating the binary question for that dimension; solution guidance and a domain-knowledge block that delimit, with Peruvian-context vocabulary, what is to count as economic, political, or uncertainty-laden content; an exception-handling block instructing the model to read tweets in any language and to answer without explanation; and an output-formatting block that constrains the reply to a single 0 or 1. Figure 3 shows how the three prompts combine: the same tweet is passed in parallel to the economic, policy, and uncertainty prompts, and the large language model returns the triple (E, P, U). For the tweet illustrated there, the model returns $(0, 1, 1)$ (no explicit economic content, but political content and an expression of uncertainty) so the tweet is counted in the political-uncertainty cell rather than the strict EPU cell.

We adopt DeepSeek-V3 for three practical reasons: strong multilingual instruction-following performance, including Spanish; an open-weights release that makes the classification step replicable;

and an inference cost low enough to label the full corpus along the three dimensions (approximately 160 United States dollars (USD)) a small fraction of the cost of comparable proprietary models at the time of classification.

Table 2 reports the eight possible (E, P, U) label combinations with their frequencies, and Table 3 summarizes the cleaned sample. Of the 213 249 tweets posted by the 97 accounts between January 2018 and January 2023, about 38% are unrelated to economics, politics, or uncertainty, while close to half carry political content. Uncertainty in Peruvian elite discourse is overwhelmingly political rather than economic: the political-uncertainty cell $(0, 1, 1)$ alone contains 59 978 tweets, against only 1 318 tagged as purely economic uncertainty $(1, 0, 1)$. The strict EPU label $(1, 1, 1)$, which requires the simultaneous presence of all three dimensions, is comparatively rare—16 451 tweets, 7.7% of the corpus—and this sparsity is precisely what motivates the complementary political-uncertainty index built from the $P \cap U$ cells.

The strict EPU count $Y_t = \sum_{i=1}^{97} \sum_{j=1}^{n_{it}} \mathbf{1}\{E_{ijt} = 1, P_{ijt} = 1, U_{ijt} = 1\}$ is positive on 1 462 of the 1 835 days, with a conditional mean of 11.25 tweets per active day. Daily tweet volume N_t is strongly right-skewed: its mean (116.21) far exceeds its median (56), and the maximum is 1 016 tweets in a single day, reflecting sharp bursts of activity around political–economic events. The index is thus well populated yet concentrated on event-driven spikes, consistent with the state-dependent behavior documented below.

3 The Index

We build a single daily index. Because every step of its construction is a choice (what to count, how to normalize it, how much to smooth it, and how to scale it) we make those choices, and the reasons for them, explicit. Each is defended below on statistical and substantive grounds, and its effect is shown on the data.

For each calendar day t, let $N_t = \sum_{i=1}^{97} n_{it}$ be the number of classified tweets posted that day and let Y_t be the number that are economic, political, and uncertain at the same time,

$$Y_t \;=\; \sum_{i=1}^{97} \sum_{j=1}^{n_{it}} \mathbf{1}\left\{E_{ijt} = 1,\; P_{ijt} = 1,\; U_{ijt} = 1\right\}, \tag{1}$$

where $E_{ijt}, P_{ijt}, U_{ijt} \in \{0, 1\}$ are the economic, political, and uncertainty labels of the j-th tweet posted by account i on day t, and n_{it} is the number of tweets that account posted that day. The two broader cells that drop one requirement are defined over the same index set,

$$Y_t^{E\cap U} = \sum_{i=1}^{97} \sum_{j=1}^{n_{it}} \mathbf{1}\left\{E_{ijt} = 1,\, U_{ijt} = 1\right\}, \qquad Y_t^{P\cap U} = \sum_{i=1}^{97} \sum_{j=1}^{n_{it}} \mathbf{1}\left\{P_{ijt} = 1,\, U_{ijt} = 1\right\}.$$

Figure 4 plots these three daily counts. Two things stand out. First, the strict count Y_t (bottom panel) and the economic-uncertainty count $Y_t^{E\cap U}$ (top panel) are almost indistinguishable in level and shape: over the sample the strict cell holds 16 451 tweets and $E\cap U$ holds 17 769, and 93 percent of the latter are also political, so the two are nearly the same object. The political-uncertainty count $Y_t^{P\cap U}$ (middle panel) is an order of magnitude larger (76 429 tweets, hundreds on a busy day against tens for the other two) and moves on its own schedule, spiking on episodes of purely political turbulence that carry no economic content. This is why the headline counts only the strict intersection: relaxing the economic requirement would swap in the much larger, differently-timed political signal. Second, all three counts are spiky and drift upward through the sample as these accounts post more, so a raw count confounds uncertainty with sheer activity.

We therefore measure the share of the day's discourse that is uncertain, normalizing Y_t by N_t,

$$s_t \;=\; \frac{Y_t}{N_t} \qquad (0 \le s_t \le 1), \tag{2}$$

and likewise for the two broader cells. Figure 5 plots these shares. The mechanical volume component is gone (the upward drift of the counts flattens) but the daily series is now noisy and ill-behaved on thin days: when only a handful of tweets are posted the denominator N_t is tiny, so a single tweet moves s_t by a large amount and the share can jump close to 1. Daily volume averages 116 tweets but its median is only 56, so thin days are common, and the early, low-volume part of the sample (2018) is correspondingly jagged.

To control that small-sample noise we pool counts over a trailing window of W days and take the ratio of the summed counts (a volume-weighted rate) rather than the average of the daily shares,

$$\rho_t \;=\; \frac{\sum_{s=t-W+1}^{t} Y_s}{\sum_{s=t-W+1}^{t} N_s}. \tag{3}$$

Pooling counts is not the same as averaging the daily shares of Eq. (2). Writing the window total $\mathcal{N}_t = \sum_{s=t-W+1}^{t} N_s$, a short rearrangement of Eq. (3) gives

$$\rho_t \;=\; \sum_{s=t-W+1}^{t} \frac{N_s}{\mathcal{N}_t}\, s_s \;=\; \sum_{s=t-W+1}^{t} w_s\, s_s, \qquad w_s = \frac{N_s}{\mathcal{N}_t}, \quad \sum_{s=t-W+1}^{t} w_s = 1, \tag{4}$$

so ρ_t is the mean of the daily shares s_s weighted by daily volume N_s. A high-volume day contributes in proportion to how much it actually says, and a low-volume day cannot dominate the window. For $t < W$ the sums in Eq. (3) run over the available days $s = 1, \dots, t$, so the first $W - 1$ dates use an expanding window.

We set $W = 14$ days. This choice reflects the trade-off between precision and responsiveness, as the data do not identify an interior optimum. Longer windows reduce the sampling noise in ρ_t, but also increase the phase lag (by $(W - 1)/2$ days for a trailing window) and smooth out the short-lived episodes the index is intended to capture. The two extremes are degenerate: minimizing noise alone drives W toward the sample length and produces a constant series, whereas minimizing lag returns the noisy daily share of Eq. (2), whose shortcomings Figure 5 displays. We adopt a fortnightly window as a middle ground between the daily frequency at which the text arrives and the monthly frequency of economic activity variables. At $W = 14$ the phase lag is 6.5 days and the sampling noise of ρ_t is a third of its standard deviation. Nothing of substance turns on the convention: recomputing the index with $W \in \{7, 21, 30\}$ leaves the main peaks, troughs, and their timing unchanged, the three series correlating with the $W = 14$ index at 0.88, 0.94, and 0.89.

Figure 6 reports that trade-off in full, one point per value of W. The horizontal axis is the phase lag, $(W-1)/2$ days. The left axis is the sampling noise of ρ_t: treating the window's tweets as draws with success rate ρ_t, its standard error is $\left(\rho_t (1-\rho_t)/\mathcal{N}_t\right)^{1/2}$, expressed relative to the standard deviation of ρ_t itself—a scale-free noise ratio, sampling error against the day-to-day variation of the rate, not a formal variance share. The right axis is the April 2020 maximum of the index, in multiples of its sample mean. Both curves fall monotonically: the noise ratio drops from 0.48 at $W = 3$ to 0.33 at $W = 14$, 0.25 at $W = 30$, and 0.19 at $W = 60$, while the April 2020 peak falls from 3.54 times the sample mean at $W = 3$ to 2.14 at $W = 60$.

Following Baker et al. (2016), we rescale the rate by its own sample mean so that the series reads in interpretable units,

$$\mathrm{EPU}_t \;=\; 100 \cdot \frac{\rho_t}{\bar{\rho}}, \qquad \bar{\rho} \;=\; \frac{1}{T}\sum_{t=1}^{T} \rho_t, \tag{5}$$

so that a value of 100 is an average day and a reading of, say, 200 is twice as uncertain as usual. On an average day $\bar{\rho} \approx 0.073$: about 7% of these accounts' discourse is economic-policy uncertainty. Rescaling is a pure change of scale, leaving the dynamics untouched.

The same two steps applied to the political-uncertainty cell $Y_t^{P\cap U}$ deliver the companion index we carry alongside the headline series. Writing $\rho_t^{P\cap U} = \sum_{s=t-W+1}^{t} Y_s^{P\cap U} / \sum_{s=t-W+1}^{t} N_s$ for the volume-weighted fortnightly rate of that cell, computed exactly as in Eq. (3) but with $Y_s^{P\cap U}$ in the numerator, the political-uncertainty index is that rate on the same base-100 scale,

$$\mathrm{PU}_t \; = \; 100 \cdot \frac{\rho_t^{P\cap U}}{\bar{\rho}^{P\cap U}}, \qquad \bar{\rho}^{P\cap U} \; = \; \frac{1}{T}\sum_{t=1}^{T} \rho_t^{P\cap U}, \tag{6}$$

so that PU_t and EPU_t are read in the same units and are directly comparable, each equal to 100 on an average day of its own cell.

Figure 7 plots the finished index against its two simpler forms, the raw count $100\, Y_t/\bar{Y}$ and the daily share $100\, s_t/\bar{s}$, each rescaled by its own mean so that all three average 100 by construction. The mean therefore cannot separate them; the median can, and the further it falls below 100 the more the average is carried by a few extreme days. The raw count has a median of 44.6 and a maximum of 915, because a handful of high-volume days carry the average; the daily share, a median of 86.3 and a maximum of 1 343, because a thin denominator lets one tweet dominate. The refined index has a median of 96.4 and a maximum of 272: its typical day sits close to the base, and its extremes are episodes rather than artifacts of volume or of a small denominator.

Figure 8 shows the index at work. The top panel is the headline EPU; the two below are the VIX and the log return of the Lima exchange, with the eight domestic crisis windows of Section 5 shaded. The index rises sharply inside the windows of economic origin, most clearly the COVID-19 emergency of March 2020, where it peaks at 2.7 times its average. The purely political ruptures (the November 2020 vacancy of President Vizcarra, the December 2022 removal of President Castillo) register instead in the companion PU index of Section 3, while the strict headline series stays near or below its base. Where the index rises, the VIX and the market returns tend to move with it. The figure cannot say how strong that co-movement is, whether it holds outside the shaded windows, or whether it survives the persistence of the three series. The sections that follow measure it, asking how closely the index resembles established gauges of uncertainty.

4 Methodology

This section assesses how closely the headline economic and political uncertainty (EPU) index of Section 3 resembles established measures of uncertainty. We ask a question of similarity: whether a text-based attention index built from Peruvian elite discourse behaves like market-based gauges of uncertainty, and where it departs from them. A close co-movement is evidence that the index tracks the same underlying phenomenon (Baker et al., 2016; Bloom, 2009); a divergence is itself informative about what each measure captures. We use a set of association measures—static and rolling correlations against the VIX and BVL realized variance, and the conditional volatility of the BVL return from a GARCH(1, 1) model, compared as a third series on the same footing with the index and the VIX.

4.1 Benchmark Indicators

We compare the EPU index against two external markets, which between them supply three benchmark series. The first market is the Chicago Board Options Exchange (CBOE), whose Volatility

Index (VIX), the implied volatility of options on the Standard and Poor's 500 (S&P 500) index, is the most widely used forward-looking gauge of global financial uncertainty. The second is the Bolsa de Valores de Lima (BVL), which enters twice, because a return is not a volatility and the two ways of turning it into one behave very differently. Let P_t denote the closing level of the BVL index on day t and $r_t = \log(P_t/P_{t-1})$ its daily log-return. The first object is the squared log-return,

$$\mathrm{RV}_t = r_t^2, \tag{7}$$

and the second is $\hat{\sigma}_{\mathrm{BVL},t}$, the conditional volatility extracted from r_t itself by the GARCH(1, 1) filter of Section 4.4. The filter is applied to the return r_t, never to r_t^2, which is already a variance proxy. The distinction matters throughout: RV_t is a one-day spike process while $\hat{\sigma}_{\mathrm{BVL},t}$ is a smooth fortnight-scale series, and Section 6.2 shows that the index is nearly orthogonal to the first and correlated with the second. This is the standard one-day proxy: if the conditional mean of r_t is zero, then r_t^2 is an unbiased, though noisy, estimate of that day's return variance.[1]

The two benchmarks differ in every dimension except one. The VIX is global, forward-looking, and read off option prices; RV_t is domestic, backward-looking, and read off observed returns. What they share is that both are anchored to traded financial instruments, unlike our text-based index. Both are quoted on trading days only, while the index is defined on every calendar day.

Both markets are quoted on trading days only, while the index is defined on every calendar day, and that mismatch has to be resolved before any statistic is computed. Every financial statistic in this paper is computed on the common trading-day panel, not on the calendar panel. The reason is a property of the market data files as they were supplied to us, and it invalidates the calendar-panel estimates of the previous version.

The three market series arrive on all 1 835 calendar dates, with the non-trading dates filled by *linear interpolation in levels* rather than by carrying the last quote forward. Two facts establish this: the sample contains no return exactly equal to zero, which a carried-forward price would produce mechanically on every non-trading date; and on every weekend the price lies on the straight line joining the Friday and Monday quotes to within the fourth-decimal rounding of the file. We identify the filled dates by that vanishing second difference and validate the set against the Peruvian public-holiday calendar: of the 77 weekday dates flagged for the BVL, 44 are recognised national holidays. Interpolation manufactures information rather than repeating it. Spreading one genuine Friday-to-Monday move evenly across three dates suppresses the realised variance on all three and induces positive autocorrelation: the daily return has a standard deviation of 0.0132 on trading dates against 0.0092 on the calendar panel, so the filled panel understates volatility by 30 percent, and its first-order autocorrelation is 0.074 against 0.004, so essentially all of the apparent return predictability in the calendar panel is an artifact of the fill. The BVL file carries a second and worse defect: between 22 January and 3 March 2021 the index is an exact arithmetic progression, rising by 2.3711 points on each of 40 consecutive dates, weekends included, with an implied return standard deviation of 0.00003 against 0.0114 for the rest of 2021. That six-week stretch is pure interpolation and no volatility model can be estimated across it; our rule removes it automatically.

We therefore keep only dates on which all three market series carry a genuine quote. The BVL has 1 234 such dates, the VIX 1 221, and the PEN/USD rate 1 273; the Peruvian and United States holiday calendars do not coincide, so their intersection is 1 143 dates, of which $T = 1\,142$ admit a return. Of those 1 142, 218 fall inside the crisis windows of Section 5 (78 economic, 140 political,

[1] Write $r_t = \sigma_t z_t$, where σ_t^2 is the day's variance and z_t has zero mean and unit variance given the past. Then $\mathbb{E}\left[r_t^2\right] = \sigma_t^2\,\mathbb{E}\left[z_t^2\right] = \sigma_t^2$, which is the unbiasedness. The estimate rests on a single observation, however: under normality z_t^2 is chi-square with one degree of freedom and variance 2, so $\mathrm{Var}\left(r_t^2\right) = 2\sigma_t^4$ and the standard deviation of the proxy is $\sqrt{2} \approx 1.41$ times the quantity it estimates.

70 political excluding C5) and 924 are calm. The index itself remains defined on all 1 835 calendar days and is simply read on the trading dates; no index observation is interpolated. Figure 8 plots EPU_t, VIX_t and RV_t together.

4.2 Static Co-Movement

For each benchmark $b_t \in \{\mathrm{VIX}_t,\ \mathrm{RV}_t\}$ we report two association statistics between the EPU index E_t and b_t over a sample of T days. The Pearson correlation measures linear co-movement,

$$\rho \;=\; \frac{\sum_{t=1}^{T}(E_t-\bar{E})(b_t-\bar{b})}{\sqrt{\sum_{t=1}^{T}(E_t-\bar{E})^2}\,\sqrt{\sum_{t=1}^{T}(b_t-\bar{b})^2}}, \tag{8}$$

with $\bar{E}$ and $\bar{b}$ the sample means. The Spearman correlation is the Pearson correlation of the ranks: with $R_t = \mathrm{rank}(E_t)$ and $Q_t = \mathrm{rank}(b_t)$,

$$\rho_S \;=\; \frac{\sum_{t=1}^{T}(R_t-\bar{R})(Q_t-\bar{Q})}{\sqrt{\sum_{t=1}^{T}(R_t-\bar{R})^2}\,\sqrt{\sum_{t=1}^{T}(Q_t-\bar{Q})^2}}, \tag{9}$$

which captures any monotone relation and is robust to the heavy tails of uncertainty series. Together they discriminate the shape of the association: $\rho_S \approx \rho$ signals a linear link, $\rho_S > \rho$ signals a monotone but nonlinear one, and $\rho_S < \rho$ signals that a few jointly extreme episodes are lifting the linear correlation above the rank correlation, which is the case here (0.426 against 0.429). We use ρ_S only to diagnose the shape of the association and conduct inference on ρ. The reason is practical: ρ can be written as the slope of a least squares regression, so its standard error follows from the long-run variance derived in Appendix A.2, whereas no comparable correction for serial dependence is available for a rank correlation.

Before measuring co-movement we classify each series under study (the EPU index, the VIX, BVL realized variance, and the filtered volatility $\hat{\sigma}_{\mathrm{BVL},t}$), generically denoted x_t, by how persistent it is. Two things hang on the answer: a series that does not revert to a fixed mean would have to enter in first differences, since a correlation in levels would measure shared drift; and the more persistent the series, the less each additional day adds, which the standard errors must correct for. Table 4 reports the diagnostics on the $T = 1\,142$ trading-day panel of Section 4.1, the same panel every other financial statistic in the paper uses. This is worth stating twice, because the two panels give different answers: on the interpolated 1 835-day calendar panel the same diagnostics return a first-order autocorrelation of 0.980 for the VIX against 0.964 here, and the fill is what creates the difference.

Two tests with opposite nulls are needed, because neither settles the question alone. The augmented Dickey–Fuller (ADF) test of Dickey and Fuller (1979), augmented as in Said and Dickey (1984), regresses the change of the series on its own lagged level and asks whether deviations are pulled back toward the mean; its null is a unit root, rejected below -2.86 at 5%.[2] The Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test of Kwiatkowski et al. (1992) reverses the null: it tests whether the series contains a random-walk component, through the accumulated deviations from its mean, which stay bounded under stationarity and grow without bound under a unit root. It rejects above 0.463 at 5%. Appendix A.1 states both in full.

The two verdicts are jointly informative, and Table 4 delivers them. The EPU index and the VIX reject the unit root (-3.84, -3.88) and also reject stationarity (1.726, 3.775). The two failures

[2] $\Delta x_t = c + \pi x_{t-1} + \sum_k \delta_k \Delta x_{t-k} + u_t$: a unit root is $\pi = 0$, no fixed level to return to, while $\pi < 0$ undoes a fraction $|\pi|$ of any deviation each day. The lagged differences only whiten the residual.

are of different kinds: the augmented Dickey–Fuller test has low power against processes near the unit-root boundary, whereas the KPSS statistic over-rejects in finite samples when the series is stationary but strongly persistent. With $T = 1\,142$ the first test is powerful enough to detect that the autoregressive root lies below unity, while the second is distorted enough to reject stationarity, and that combination is the signature of the near-unit-root region: first-order autocorrelations of 0.967 and 0.964, and a shock to the level with a half-life of a few weeks, 20.6 days for the index and 18.8 for the VIX. Both series are highly persistent before any benchmark is brought in, the index marginally more so than the VIX, decaying over a horizon of weeks rather than days. This pattern is also consistent with structural breaks or regime shifts, so we read it as descriptive rather than as uniquely justifying the levels specification. BVL realized variance behaves differently, reverting within a day with an autocorrelation of 0.168, so its shocks are gone before the index has begun to respond to them. Its KPSS statistic (1.086) also exceeds the critical value, for an unrelated reason: the mean of RV_t rises from 4.4×10^{-5} before 2020 to 2.6×10^{-4} afterwards, and that shift in level, not serial dependence, is what inflates the partial sums. The filtered volatility $\hat{\sigma}_{\mathrm{BVL},t}$ sits at the other extreme, with an autocorrelation of 0.987 and a half-life of 54.7 days: the GARCH filter turns the same return that produces RV_t into the most persistent series in the table, which is exactly why it, and not RV_t, is comparable to a fortnightly index.

The persistence of E_t and of the VIX bears directly on inference. The classical standard error assumes each day contributes fresh information, and here it does not: the index is a fourteen-day rolling rate, so consecutive observations share thirteen days of tweets, and the VIX inherits the persistence of the option market. Appendix A.2 writes $\hat{\rho}$ as a least squares slope and shows that its sampling error is governed by the long-run variance of the score, Eq. (A.2), which the classical formula understates by a factor of up to three and a half for the VIX and by almost nothing for RV_t. We estimate it with the heteroskedasticity- and autocorrelation-consistent (HAC) estimator of Newey and West (1987), weighted by the triangular window of Bartlett (1950), which guarantees a nonnegative estimate. The bandwidth follows the plug-in of Andrews (1991) applied to $\hat{g}_t$, long for the VIX and short for RV_t; we report $L \in \{7, 30, 60, 100\}$, the lower end being the sample-size rule $L = \left\lfloor 4\,(T/100)^{2/9} \right\rfloor$ of Newey and West (1994), which returns 7 on the 1 835-day calendar panel and 6 on the $T = 1\,142$ trading-day panel used here; the two differ only in the third decimal of the standard error (0.066 against 0.069 in the full-sample VIX row), and we keep 7 so that the column is comparable with the previous version. The exponent 2/9 is not a free choice, and Appendix A.2 says where it comes from. For reference we also report the classical standard error $\left(1 - \hat{\rho}^2\right)/T^{1/2}$.

Under independence the sample autocovariances of the fitted score vanish, $\hat{\gamma}_j = 0$ for $j \geq 1$, Eq. (A.4) collapses to $\hat{\gamma}_0/T$, and for jointly Gaussian data this is the classical variance $\left(1 - \rho^2\right)^2/T$ of a Pearson correlation. That collapse fails here. The index is a fourteen-day rolling proportion, so consecutive observations share thirteen of their fourteen days of tweets by construction, and the VIX inherits the persistence of the option market. The fitted scores are therefore autocorrelated: at first order 0.86 for the VIX against 0.13 for RV_t. The neglected $\hat{\gamma}_j$ are large and positive, so the classical formula understates the standard error of $\hat{\rho}$. Section 6.1 reports both. The correction is large for the VIX and small for RV_t, matching their first-order autocorrelations.

Each correlation is computed on the full sample and, separately, on crisis and calm days using the eight windows of Section 5. Dated from the historical record rather than from market movements, the split is exogenous to the VIX and to RV_t and cannot be an artefact of the benchmarks under evaluation. A correlation computed on the pooled crisis days measures whether the index and the benchmark are high in the same episodes. It does not measure whether they move together day by day inside one: within C3 alone the correlation is -0.033, though both series sit far above their sample means throughout the window. Joint elevation across episodes is the property we ask of the index, and it is the one this split can establish. Pooling also absorbs the difference between the two regime means, so the full-sample value need not lie between the crisis and calm values. Each episode has a known date: the resignation, the dissolution, the vote. What we choose is where the window around it begins and ends. We therefore move every start and end date by -5, 0, and $+5$ days, half the length of the shortest window, and report the range of the resulting correlations.

4.3 Time-Varying Co-Movement

The crisis/calm split treats similarity as constant within each regime. To let it vary over time we compute the Pearson correlation of Eq. (8) on a trailing window of $m = 90$ days. The window length trades sampling noise against time resolution: the standard error of a correlation falls roughly like $m^{-1/2}$, so a shorter window tracks change faster but at each date estimates ρ from fewer observations and is noisier, while a longer one is smoother but blends distinct episodes and lags turning points by about $m/2$ days. We set $m = 90$ days, one trading quarter. Because the index is a 14-day rate and both series are highly persistent, the effective number of independent observations is well below 90, so we read the rolling curve as descriptive rather than as precise local inference. Hence, we define

$$\rho_t^{(m)} \;=\; \frac{\sum_{s=t-m+1}^{t}(E_s-\bar{E}_t)(b_s-\bar{b}_t)}{\sqrt{\sum_{s=t-m+1}^{t}(E_s-\bar{E}_t)^2}\,\sqrt{\sum_{s=t-m+1}^{t}(b_s-\bar{b}_t)^2}}, \tag{10}$$

where $\bar{E}_t$ and $\bar{b}_t$ are the means of E_s and b_s over the window $s = t-m+1,\dots,t$. A quarter-long window resolves only the two crises longer than 100 days, the pandemic (C3) and the contested election (C5); the six shorter windows are averaged into the surrounding calm and are left to the discrete split of Section 4.2.

Figure 9 plots $\rho_t^{(90)}$ for the VIX and for RV_t with the crisis windows shaded, and three features of each curve are informative. The level of a curve away from the shaded bands measures the similarity in ordinary times; its height inside the bands, relative to that level, measures how much the similarity changes in crises. The sign locates disagreement: an interval where $\rho_t^{(90)} < 0$ is one in which the index and the benchmark moved in opposite directions. And the alignment of any rise with a shaded band distinguishes co-movement tied to a documented crisis from a rise outside every band, which flags an episode the narrative dating did not mark. The curve is read as the time profile of the similarity.

4.4 Similarity in Conditional Volatility

The VIX is already a volatility, the option-implied volatility of the S&P 500, so its level is directly comparable to the level of the index. The BVL return is not a volatility; it is the daily price change, and what plays the role of a volatility is its dispersion over time. To place the Lima market on the same footing as the VIX we extract its conditional volatility with a GARCH$(1,1)$ filter, which turns the return into a volatility series comparable, as a level, with the VIX and with the index. We fit the same filter to the index and to the VIX as well, but for a different purpose: to report the diagnostics of Table 5 and to establish, in Section 6.2, why the conditional variance of the index measures how precisely its rate is estimated rather than how much uncertainty there is, so that it is the level of the index, and not a volatility of its own, that carries the comparison. The filter also matches the two objects in horizon: with $\hat{\beta} = 0.940$ for the BVL return on the trading-day panel, its effective memory is $1/\left(1-\hat{\beta}\right) = 16.6$ days, the same fortnightly order as the $W = 14$ window over which the index is pooled, and that is the quantitative reason the association with the Lima market rises from 0.061 against the one-day realized variance to 0.453 against the filtered volatility.

Let x_t denote the series being modelled and let $\mathcal{F}_{t-1} = \sigma(x_{t-1}, x_{t-2}, \ldots)$ denote the information set available at the end of day $t-1$; all conditional moments are taken with respect to it. The object of interest is the conditional variance $\sigma_t^2 = \mathrm{Var}\,(x_t \mid \mathcal{F}_{t-1})$ of Eq. (A.5), the day-by-day counterpart of the unconditional variance $\mathrm{Var}(x_t)$: its square root σ_t, the conditional volatility, is the daily measure of turbulence the model delivers.

The model is the generalized autoregressive conditional heteroskedasticity model of order $(1, 1)$, GARCH$(1, 1)$, of Bollerslev (1986). We impose two restrictions on it: the innovation is Gaussian, and the variance responds to the magnitude of yesterday's innovation and not to its sign. It generalizes the autoregressive conditional heteroskedasticity (ARCH) model of Engle (1982), in which today's variance responds only to recent squared shocks; adding yesterday's variance itself as a regressor lets a single lag of each capture the long, smooth decay of volatility that would otherwise require many ARCH lags.

The mean equation is a constant for the return series r_t and adds a first-order autoregressive, AR(1), term for the level series (the VIX and the EPU index), so that in every case the GARCH recursion describes the conditional volatility of a mean-zero innovation:

$$x_t = \mu + \phi\, x_{t-1} + \varepsilon_t, \qquad \varepsilon_t = \sigma_t\, z_t, \qquad z_t \sim \mathrm{N}(0, 1) \text{ i.i.d.}, \tag{11}$$

$$\sigma_t^2 = \omega + \alpha\, \varepsilon_{t-1}^2 + \beta\, \sigma_{t-1}^2, \qquad \omega > 0, \quad \alpha \geq 0, \quad \beta \geq 0, \quad \alpha + \beta < 1, \tag{12}$$

where μ is the intercept and ϕ the AR(1) coefficient, set to zero for the return series. The innovation $\varepsilon_t \equiv x_t - \mathbb{E}[x_t \mid \mathcal{F}_{t-1}]$ is a martingale-difference sequence, $\mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0$ and $\mathbb{E}[\varepsilon_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2$: the sign of today's surprise is unpredictable, its expected magnitude is not, and that predictability of magnitudes is what the variance equation models. In Eq. (12), ω is the variance intercept, α governs the response of today's variance to yesterday's squared innovation (the news term), and β the persistence inherited from yesterday's variance (the memory term).

The recursion of Eq. (12) is initialized at σ_0^2 and, by induction, σ_t^2 is a measurable function of $x_{t-1}, \ldots, x_0$: it is known at $t-1$, as a conditional variance must be. Its standard properties—the long-run variance of Eq. (A.6), which is why $\omega > 0$ and $\alpha + \beta < 1$ are imposed; the geometric decay of the h-step variance forecast at rate $\alpha + \beta$, with the implied half-life $\ln(1/2)/\ln(\alpha + \beta)$ reported in Table 5; and the ARCH(∞) representation that makes the $(1, 1)$ order sufficient in practice—are collected in Appendix A.3, together with the Gaussian log-likelihood Eq. (A.7) by which the model is estimated. The estimator $\hat{\theta} = (\hat{\mu}, \hat{\phi}, \hat{\omega}, \hat{\alpha}, \hat{\beta})$ maximizes that likelihood; no closed form exists, and the maximization is carried out numerically from several starting points.

Estimation delivers a daily volatility series, which is the object we use. Given $\hat{\theta}$, the fitted innovations are $\hat{\varepsilon}_t = x_t - \hat{\mu} - \hat{\phi}\, x_{t-1}$, and the variance recursion of Eq. (12), evaluated at $\hat{\theta}$, is iterated forward through the sample,

$$\hat{\sigma}_t^2 \;=\; \hat{\omega} + \hat{\alpha}\, \hat{\varepsilon}_{t-1}^2 + \hat{\beta}\, \hat{\sigma}_{t-1}^2, \qquad t = 2, \ldots, T, \tag{13}$$

and its square root $\hat{\sigma}_t$ is the filtered conditional volatility: one number per day, in the units of the underlying series, summarizing how turbulent that day was expected to be given the information available the day before. The standardized residuals $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$ provide the model's own specification check: if the filter has absorbed the volatility clustering, the squared standardized residuals $\hat{z}_t^2$ should display no remaining autocorrelation, and we report their first-order autocorrelation for every fitted model. We fit the same GARCH$(1, 1)$ variance equation to the three series, with an AR(1) conditional mean for the two level series (the index and the VIX) and a constant mean for the BVL return; the three filtered volatilities $\hat{\sigma}_t$ are placed on a common footing by standardizing each to zero mean and unit variance.

The EPU index and the VIX are persistent levels. Their fitted persistence $\hat{\alpha} + \hat{\beta}$ measures how slowly a shock to the conditional variance dies out: the unconditional variance $\omega/(1 - \alpha - \beta)$ exists only when this sum is below one, and at the boundary $\alpha + \beta = 1$ the process is the integrated GARCH (IGARCH) of Engle and Bollerslev (1986), in which shocks never decay and no long-run variance is defined. Our estimates lie below that boundary, though close to it, so the fitted variances mean-revert slowly.

We estimate the model in levels, with the AR(1) mean of Eq. (11). The correlations reported throughout, the parameters (ϕ, α, β), and the standardized residuals $\hat{z}_t$ are invariant to positive affine re-standardizations $x_t \mapsto a + b\,x_t$ with $b > 0$, so the results do not depend on the base-100 convention of Eq. (5). Only $\hat{\mu}$, $\hat{\omega}$, and the units of $\hat{\sigma}_t$ change with the scale.

Section 6.2 reports, for each series, the estimates $(\hat{\mu}, \hat{\phi}, \hat{\omega}, \hat{\alpha}, \hat{\beta})$, the persistence $\hat{\alpha} + \hat{\beta}$, the maximized log-likelihood, and the first-order autocorrelation of $\hat{z}_t^2$, and then places three levels on one plane: the index, the VIX, and the conditional volatility $\hat{\sigma}_t$ of the BVL return, each standardized so that series measured in different units share a panel, against the crisis windows of Section 5. Pearson correlations among the three levels, on the full sample and within the crisis and calm subsamples, complete the comparison. We ask whether the three conditional-volatility series cluster in the same episodes; Section 6.2 shows that the two market series do, while that of the index does not, for a reason that is itself informative about the instrument.

4.5 What the Index May Add and Its Limitations

The three tools speak to convergent validity: agreement with the VIX and with BVL realized volatility would support the claim that the index measures uncertainty. Perfect agreement is neither expected nor desirable. The index is domestic and political, built from the discourse of Peruvian politicians, journalists, economists, and analysts, so it registers events (presidential vacancy votes, cabinet collapses, contested elections) that a United States equity-option index reflects weakly and that domestic volatility may reflect only with delay. We document where the index moves while the benchmarks do not, and the reverse.

Table 6 sets these differences out. Beyond the domestic and political focus, the measures differ in orientation and anchoring: the index reads uncertainty as expressed in discourse, the VIX as the market's expectation of future volatility, and BVL realized volatility as a summary of past returns, with only the benchmarks tied to a traded price. This is why we expect moderate rather than perfect co-movement, and why the index is not redundant with either benchmark.

Both comparisons separate crisis days from calm ones, on the expectation that a measure of uncertainty tracks the benchmarks most closely when uncertainty is high. The split rests on a calendar fixed in advance, dated from the historical record rather than from the benchmarks it evaluates. Section 5 sets out these windows for Peru over 2018–2023, including the pandemic.

5 Crisis Periods in Peru, 2018–2023

The crisis/calm split of Section 4.2 requires an explicit list of crisis windows. We date them from the documented political, institutional, and macroeconomic record (following Romer and Romer (2010) and narrative event-based risk indices (Caldara and Iacoviello, 2022)) and never from realized or implied market volatility, so the split stays exogenous to the VIX and to BVL realized volatility. An episode qualifies when it combines (i) a datable triggering event, (ii) a rupture in an institution—the presidency, Congress, the electoral authority—or a nationwide economic shock, and (iii) sustained attention rather than a one-day headline. In five years Peru had six presidents, a dissolved Congress, a contested election, an attempted self-coup, and a pandemic. Table 7 lists the eight windows.

C1. Fall of Kuczynski (March 2018). Pedro Pablo Kuczynski resigned the presidency on 21 March 2018 after the release of vote-buying recordings, and Martín Vizcarra was sworn in on 23 March. We date it 12 March–6 April 2018.

C2. Dissolution of Congress (September–October 2019). On 30 September 2019 Vizcarra dissolved Congress, which contested the act and swore in Vice President Mercedes Aráoz, who resigned within a day; the clash briefly left two claimants to the executive. We date it 27 September–15 October 2019.

C3. COVID-19 emergency (March–June 2020). The state of emergency and lockdown were declared on 15 March 2020, the sharpest output contraction in the sample and coincident with the global risk-off shock. It is the one window macroeconomic and global rather than domestic-political. We date the acute phase 15 March–30 June 2020, ending at the first major easing.

C4. Vacancy of Vizcarra and Merino interregnum (November 2020). Congress vacated Vizcarra on 9 November 2020; Manuel Merino took office on 10 November and resigned on 15 November after protests in which two demonstrators were killed; Francisco Sagasti was sworn in on 17 November. Three presidents in one week. We date it 9–18 November 2020.

C5. Contested general election (April–July 2021). The first round (11 April 2021) and the runoff (6 June) between Pedro Castillo and Keiko Fujimori were followed by weeks of annulment disputes before the electoral authorities; Castillo was proclaimed on 19 July and inaugurated on 28 July. We date it 8 April–28 July 2021, from the first round to the inauguration.

C6. Fuel-price protests and Lima stay-at-home order (March–April 2022). Rising fuel and fertilizer prices after the invasion of Ukraine triggered a nationwide transport strike. Near midnight on 4 April 2022 the government issued Supreme Decree 034-2022-PCM, ordering a 22-hour mandatory stay-at-home period for Lima and Callao on 5 April; it was revoked that same afternoon amid open defiance. Economic in origin, institutional in expression. We date it 28 March–12 April 2022.

C7. Castillo self-coup and removal (December 2022). On 7 December 2022 Castillo announced the dissolution of Congress; within hours he was impeached, removed, and detained, and Dina Boluarte was sworn in. With C4, this is the sharpest institutional rupture of the sample. We date it 7–20 December 2022.

C8. Post-Castillo protests (December 2022–January 2023). Castillo's removal set off sustained, at times fatal, nationwide protests demanding new elections and Boluarte's resignation, past the sample end. We date it 20 December 2022–31 January 2023, right-censored by the sample end.

The eight windows cover 347 of the 1 835 sample days (18.9%), leaving 1 488 calm days. Coverage is uneven by design: the pandemic (C3, 108 days) and the contested election (C5, 112 days) supply 63% of the window days, while the other six are 10-to-43-day shocks. Every year holds at least one window, so the split is not identified off a single episode. C7 and C8 are adjacent, sharing 20 December 2022; we keep them separate, as an elite institutional act differs from the popular mobilization that followed, and count the shared day once.

Figure 10 plots the windows against the headline EPU index (top) and the political-uncertainty (PU) index of Section 3 (bottom). The windows capture turbulence: the headline index averages 113 inside them against 97 outside, and its maximum (272 on 15 April 2020, 2.7 times the sample average) falls in C3. The response is heterogeneous. The strict index rises where the crisis is economic—the strict tweet share runs at 1.76 times its full-sample rate within C3 and 1.49 within the fuel-price window C6—and falls in the purely political ruptures, to half its rate in C4 and about two-thirds in C7 and C8, where the PU index spikes instead (mean 174 in C7, a $P \cap U$ share 1.54 times its rate). Two mechanisms drive the gap. Composition: in an institutional collapse the elite conversation turns political and uncertain without an economic frame, satisfying $P \cap U$ but not the triple condition $E \cap P \cap U$. Dilution: daily volume explodes, 691 tweets per day in C7 against a sample mean of 116, inflating the denominator of Eq. (3). The sparsity of the strict condition in political crises is what motivates carrying the PU index alongside it.

Two caveats govern how the figure is read. Because the index is a trailing 14-day rate, in short windows its local maximum sits near the window's end (15 October in C2, 18 November in C4) not on the trigger date. The largest readings outside any window, on 8–12 February 2018 (up to 208) and 10 August 2020 (188), fall in the thin-volume opening months, C1 averaging 16 tweets a day, so early-sample readings carry sampling noise. The list is therefore conservative: tension outside the windows works against the crisis/calm contrasts, which read as lower bounds. Every window edge is displaced by ±5 days and each correlation recomputed, so the split is assessed against its own dating rather than assumed exact.

6 Empirical Results

We apply the three tools of Section 4 to the headline index of Eq. (5). The index is defined on all 1 835 calendar days; every statistic involving a market series is computed on the common trading-day panel of Section 4.1. Each statistic is computed on the frozen daily series and is invariant to the base-100 convention.

6.1 Similarity Analysis

Table 8 reports the static co-movement of the index with the two benchmarks. Over the common trading-day panel the index and the VIX correlate at 0.429 (Pearson) and 0.426 (Spearman); the two statistics agree, so the association is linear and not merely ordinal. The split by regime runs in the direction a measure of uncertainty should show, $\rho^{\text{crisis}} = 0.581$ against $\rho^{\text{calm}} = 0.315$, with the Spearman values moving the same way (0.607 against 0.375). The resemblance more than doubles inside the eight crisis windows.

Inference must account for persistence. The fitted score of Eq. (A.2) has a first-order autocorrelation of 0.864 for the VIX, so the classical standard error treats highly overlapping days as fresh; the Newey–West estimator of Eq. (A.4) returns 0.076 (Andrews plug-in $L = 10$). The full-sample correlation is 5.7 HAC standard errors from zero, the crisis correlation 5.8 against a HAC standard error of 0.101, and the calm correlation 4.6 against a HAC standard error of 0.069.

The index shows no such relation with BVL realized variance $\text{RV}_t = r_t^2$: the Pearson correlation is 0.061 against a HAC standard error of 0.038, the crisis and calm values (0.021 and 0.056) are equally indistinguishable from zero, and displacing every window edge by ±5 days moves them only within $[-0.031, 0.052]$ and $[0.017, 0.085]$ respectively. The HAC correction barely moves the standard error here (0.038 against a classical 0.029), because the score autocorrelation is low rather than near unity—large exactly where the persistence is and negligible where it is not, the check promised in Section 4.2. The absence is a statement about frequency, not about the index: RV_t reverts within a day while the index is a trailing fortnightly rate, so a day on which Lima moves sharply is invisible to a two-week average. What the two might share is the smooth, persistent component of variance, which the GARCH filter of Section 6.2 recovers.

The EPU–VIX similarity is between-episode, not day-to-day, and three readings agree. Window by window the correlation ranges from -0.790 in C1 to 0.761 in C6 and is -0.033 inside the pandemic window C3. C1 and C6 hold 18 and 12 trading days, so those two extremes are mostly sampling noise rather than evidence of anything, which reinforces the reading that there is no co-movement within episodes. What lifts the pooled crisis figure to 0.581 is that C3 carries an index mean of 183.7 against a sample average of 99.1, which is 1.98 standard deviations above it, while the

VIX sits 1.95 standard deviations above its own mean over the same window, so the two are jointly elevated in the same episodes rather than tracking each other within them. The rolling correlation of Figure 9 tells the same story: it averages 0.055 over the sample, is negative on 42.8% of windows, rises to 0.260 inside the crisis bands (peaking at 0.656 in the window closing 26 August 2020, which still spans the tail of C3), and sits at 0.007 outside. In first differences the association vanishes, -0.029 over the sample, 0.003 in crises, and -0.038 in calm, all indistinguishable from zero (Figure 11). This is the signature of a measure that is not redundant with the benchmark, and it is sharpest where the two disagree: the curve falls to -0.29 around the November 2020 vacancy, when the conversation before the vote turned political without an economic frame and the index fell while the VIX rose into the United States election. The EPU–RV_t rolling curve is flat throughout (mean 0.003, negative on 49.5% of windows).

Figure 9 is where this is read directly, along the three features set out in Section 4.3. Away from the shaded bands the VIX curve swings widely about zero and spends almost half the sample below it; inside the bands it lifts, reaching its highest values across C3; and it falls to its trough around the November 2020 vacancy, while the RV_t curve drifts about zero without ever tracking the bands. What this licenses is a claim about timing rather than strength: the EPU–VIX association is episodic, switched on by documented crises, not a standing relation that merely gets stronger.

Figure 11 settles the frequency at which that association lives. Panel (a) overlays the rolling correlation in levels on the same correlation in first differences, and the thing to look at is whether the swings of the first survive in the second; Panel (b) scatters the daily change in the index against the daily change in the VIX. The differenced curve stays flat near zero through every window and the scatter is a centred cloud with no discernible slope, so the shared component is entirely between episodes: the index and the VIX rise and fall together across the sample without the index echoing the benchmark from one day to the next, which is why it cannot be read as a daily tracker of it.

A different question, and the one a user of the index would ask first, is whether it separates the days a historian would call a crisis from the days around them. Table 9 and Figure 12 answer it with the area under the receiver operating characteristic curve, the Mann–Whitney estimate of $\Pr\left(X^{\text{crisis}} > X^{\text{calm}}\right)$: 0.5 is a coin flip, values above it mean the series runs higher on crisis days, values below it the reverse. It fits no model and imposes no lag, and the labels come from Table 7, dated from event chronologies rather than from market data, so the exercise is not circular. Appendix A.4 sets out the statistic in full: it is the Mann–Whitney probability of Eq. (A.10), it depends on the data only through ranks and so is invariant to the base-100 convention and to any monotone transformation, it does not depend on how many crisis days the calendar contains, and by Eq. (A.12) a value below one half is an inverted detector carrying as much information as its mirror image.

Pooled over all 218 crisis trading days the headline index scores 0.546, barely above the coin flip. That figure is a mixture artifact. Splitting the windows by the character of the episode (recorded in Table 7 and using no information from any series) resolves it: the index scores 0.939 on the 78 trading days of the two economic windows and 0.327 on the 140 of the six political ones. Panel A shows the pattern window by window, 0.962 for the COVID-19 emergency C3 and 0.813 for the fuel-price episode C6 against 0.101 for the Merino interregnum C4 and 0.257 for the post-Castillo protests C8. The index is close to a perfect detector of the economic windows and an inverted one of the political windows.

The design deserves a word here, because it had every condition to be trivially confirmatory: an index built from the discourse of an elite that reacts to political and economic events is being scored against windows dated from those same events, so a uniformly high reading would have proved very little. What makes the exercise informative rather than circular is precisely where the index fails—the 0.101 of C4, far below the coin flip, is a number no circular design would produce, and it shows the series is separating episodes by their economic content instead of echoing the calendar it is scored against.

Figure 12 shows the same result in the form the statistic comes from. Panel (a) pools the economic windows and Panel (b) the political ones against the same calm days, and what to look at is which side of the diagonal each curve falls on: the headline index runs above it in Panel (a) and below it in Panel (b), while the political-uncertainty index runs above it in both. A curve below the diagonal is not a failed detector but an inverted one, carrying as much information as its mirror image, which is the graphical form of the claim that the two indices divide the crisis calendar rather than compete over it.

This is a property of the construction rather than a defect of it. The headline index counts a tweet only in the $E \cap P \cap U$ cell of Section 3: a constitutional crisis with no immediate economic content does not qualify, and the fortnightly denominator falls as political traffic that fails the economic test crowds the timelines. The political-uncertainty index of Eq. (6), which requires only $P \cap U$, is the mirror image, and its pooled score of 0.695 on the political windows understates it for the same reason: the contested general election C5 spans 70 of the 140 political trading days and returns 0.497, so half that pooled sample is a single window no series in the table separates. Dropping C5 leaves 70 political trading days on which the political index scores 0.893 against 0.364 for the headline index, a difference of 0.529 with a DeLong z of 16.69 and a block-bootstrap interval of [0.340, 0.689]; on the economic windows the same difference is -0.234, interval $[-0.350, -0.097]$. The DeLong statistic assumes independent observations, which the persistence documented in Table 4 rules out here, so the block-bootstrap interval is the valid number and the z is reported only as a conventional reference. The sign reverses with the character of the episode and the two scores are near mirrors, 0.939 for the headline index on economic days and 0.893 for the political index on political days. The indices are not competing measures of one quantity; they partition the crisis calendar between them.

The benchmarks place this reading. The VIX scores 0.868 on the economic windows and 0.509 on the political ones, the coin flip, rising only to 0.535 once C5 is dropped, against 0.893 for the political index: it registers the pandemic because the pandemic was global and is close to blind to every domestic political episode in the sample. Realized variance scores near 0.59 on all windows and 0.531 without C5, discriminating weakly and indiscriminately. No series available off the shelf separates Peruvian political crises from calm days, and this is the gap the index family of this paper fills. C5 itself, 70 trading days that average an acute fortnight together with three quiet months, resists every series in Table 9, which is a property of the coarse dating of Table 7 rather than of the indices.

6.2 Conditional Volatility

Table 5 reports the GARCH$(1,1)$ estimates for the three series. The VIX has a large news coefficient and a moderate memory coefficient, $\hat{\alpha} = 0.461$ and $\hat{\beta} = 0.536$: implied volatility reprices sharply and carries little memory in the filter, whose effective length $1/\left(1 - \hat{\beta}\right)$ is only 2.2 days. Its persistence $\hat{\alpha} + \hat{\beta} = 0.997$ is nonetheless close enough to unity that the implied half-life of 243.9 days in Table 5 should be read as a statement about the boundary rather than as a forecast horizon. The BVL return sits at the opposite corner, $\hat{\alpha} = 0.060$ and $\hat{\beta} = 0.940$, the familiar equity configuration in which the variance carries almost all the memory; its persistence reaches the numerical boundary, so close to the integrated case that the implied half-life describes how slowly the filter forgets rather than a horizon anyone would forecast over. The index is nearer that corner, $\hat{\alpha} = 0.085$ and $\hat{\beta} = 0.886$, with persistence 0.9708, and its mean coefficient $\hat{\phi} = 0.974$ reproduces the near-unit-root persistence Table 4 found in the level; its half-life carries the same caution. The filter absorbs the volatility clustering: the first-order autocorrelation of the squared standardized residuals is 0.015, -0.008, and 0.046 for the index, the VIX, and the BVL return, and the Ljung–Box statistic of Ljung and Box (1978) at ten lags gives p-values of 0.24, 1.00, and 0.75. The 0.06 for the BVL return is marginal evidence and nothing more: it does not reject at the 5% level, and we do not read it as significant, nor as establishing that the filter is adequate. It is reported so that the reader can weigh it. On the trading-day panel the same statistic returns $p = 0.75$ under the Gaussian filter and $p = 0.89$ under Student-t errors, so the marginal value in the previous version was itself an artifact of the interpolated dates rather than a property of the return. Two caveats attach. The mean equation is deliberately minimal, so the level residuals retain autocorrelation (a fourteen-day rate is a moving average of order thirteen, which an AR(1) cannot span); and the standardized residuals are far from Gaussian, so Eq. (A.7) is a quasi-likelihood, consistent for (ϕ, α, β) under Eqs. (11)–(12) even when the innovation is not normal (Bollerslev and Wooldridge, 1992).

Figure 13 plots the four series of this comparison on the trading-day panel, and Figure 14 shows what interpolating the non-trading dates does to the filter itself. Figure 15 places the three series on one standardized scale: the index, the VIX, and the conditional volatility $\hat{\sigma}_{\text{BVL}}$. All three rise together in the crisis windows, most sharply in the pandemic window C3, and settle together between them, the index moving with both market measures but with wider swings and looser timing, as a domestic text-based measure and a global financial one should. Table 10 confirms it. The two market measures agree most, at 0.684 over the sample, which shows the GARCH filter recovers a genuine volatility; the index sits alongside both, at 0.429 with the VIX and 0.453 with $\hat{\sigma}_{\text{BVL}}$, the latter stable across regimes (0.468 in crises, 0.424 in calm). The correlations are moderate, between 0.43 and 0.45 for the index: the co-movement is real and the series share a common uncertainty component, while each carries a dimension the others do not.

Figure 16 explains why the index enters through its level and not a volatility of its own. The index is a rate estimated on a finite number of tweets, with sampling error $\left(\rho_t\left(1-\rho_t\right)/\mathcal{N}_t\right)^{1/2}$ for $\mathcal{N}_t$ the trailing fourteen-day tweet volume, so a day-to-day movement is part signal and part measurement noise, and the noise shrinks as the platform gets busier. Regressing $\ln \hat{\sigma}_t^2$ on $\ln \mathcal{N}_t$ returns a log-log elasticity of -0.296 with $R^2 = 0.438$, and $\hat{\sigma}_t$ correlates at 0.675 with $\mathcal{N}_t^{-1/2}$; a pure binomial sampling term would deliver a slope of -1, so a substantial part of the variation in the index's conditional variance tracks the precision of the measurement rather than turbulence in the underlying rate. Because volume is procyclical to crises (162.1 tweets a day inside the windows against 115.2 outside), the index is measured most precisely exactly when the market measures are most turbulent: the two second moments are pushed in opposite directions by construction. The variance of EPU_t is therefore a measurement object, the similarity with the benchmarks lives in the first moment, and a variance equation augmented with volume would model the instrument rather than the uncertainty it measures. We report the symmetric Gaussian GARCH$(1,1)$ and stop there.

6.3 What the Index Contributes

The evidence supports a bounded claim. The index is a daily, interpretable, and replicable series of economic and political uncertainty for Peru, built from public platform text and a documented classification, where no comparable instrument existed. Its full-sample correlation is 0.429 with the VIX and 0.453 with Lima market volatility, rising to 0.581 with the VIX in crises: close enough to place it among established uncertainty measures, far enough not to be a copy of any. What it adds is domestic and narrative. The VIX is a global financial gauge and BVL volatility a market reaction; neither registers a presidential vacancy vote, a cabinet collapse, or a contested election as an uncertainty event in its own right. The index does, and the episodes on which it and the

VIX diverge, C4 and C7 above all, are exactly the domestic institutional ruptures a United States equity-option index has no reason to price. Paired with the political-uncertainty index of Eq. (6), it separates economic-policy uncertainty from political rupture. We do not claim the index is a better measure of uncertainty in general, but that it is the more appropriate instrument for domestic economic and political uncertainty in Peru, while the VIX and BVL volatility capture the global-financial dimension the index is not built to see.

7 Conclusions

We have built the first daily index of economic and political uncertainty for Peru from the X discourse of 97 influential accounts, classifying 213 249 tweets along economic, political, and uncertainty dimensions with a large language model. The index is the volume-weighted fortnightly rate of Eq. (3), rescaled to base 100 by Eq. (5).

The result we would put first is that the index identifies crises. Scored by the area under the receiver operating characteristic curve against the eight windows of Table 7, dated from the historical record and so exogenous to every series compared, the headline index reaches 0.939 on the two economic episodes and 0.962 on the pandemic window alone, against 0.868 and 0.919 for the VIX and 0.610 and 0.642 for realized variance. On the six political ruptures it runs below the coin flip, at 0.327, which by Eq. (A.12) is an inverted detector rather than a failed one: the strict $E \cap P \cap U$ cell requires an economic frame that an institutional collapse does not supply, and the fortnightly denominator swells as political traffic crowds the timelines. The companion political-uncertainty index is the mirror, reaching 0.893 on the political windows once the contested election C5 is set aside, against 0.535 for the VIX and 0.531 for realized variance. The paired difference is +0.529 with a block-bootstrap interval of [0.340, 0.689]. Neither benchmark registers a presidential vacancy vote or a self-coup as an uncertainty event, and that is the gap this index family fills.

In levels the index also tracks the VIX, and the association is state-dependent: a full-sample Pearson correlation of 0.429 rises to 0.581 in the 218 crisis trading days and falls to 0.315 in the 924 calm ones, an ordering that survives Newey–West inference at every bandwidth and a ±5-day shift of every crisis boundary. It does not track the one-day realized variance of the BVL (correlation 0.061, 1.6 HAC standard errors from zero), because a squared daily return reverts within a day while the index averages a fortnight. Filtered through a symmetric Gaussian GARCH$(1, 1)$, the conditional volatility of the BVL return enters as a third measure, correlating 0.684 with the VIX and 0.453 with the index: the two market volatilities agree most, confirming the filter recovers a genuine volatility. We do not add the index's own conditional volatility as a fourth series, because it is a measurement object: regressing $\ln \hat{\sigma}_t^2$ on $\ln \mathcal{N}_t$ gives a slope of -0.296 ($R^2 = 0.438$), so tweet volume, which rises in crises, makes the index most precise exactly when the benchmarks are most turbulent. The index carries its signal in the first moment (Figure 17).

The correlations lie below unity, so the index is not a rescaling of either benchmark, and above zero, so the three share a common component: the index is not redundant. What it adds is domestic and low-frequency. The VIX similarity is between-episode, not daily (in first differences the correlation is -0.029), and the episodes where the two diverge, C4 and C7, are the domestic institutional ruptures a United States equity-option index has no reason to price; there the headline index falls below its base while the political-uncertainty index built from $P \cap U$ rises to a mean of 174. The two indices are not competing measures of one quantity. They partition the crisis calendar, and the discrimination exercise is where that partition is visible.

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A Econometric Background

This appendix collects the standard material that Section 4 uses and does not need to derive: the two tests with opposite nulls behind Table 4, the heteroskedasticity- and autocorrelation-consistent variance of a correlation, and the properties and estimation of the GARCH(1, 1) model behind Table 5. None of it is new. It is set out here so that the main text can state what is done with it, and with what consequence for this index, without interruption.

A.1 Unit-Root and Stationarity Tests

Two tests with opposite nulls are needed, because neither settles the question alone. The augmented Dickey–Fuller (ADF) test of Dickey and Fuller (1979), in the augmented form of Said and Dickey (1984), regresses Δx_t on x_{t-1}, a constant, and enough lagged differences to whiten the residual; its null is a unit root—a random-walk-like process with no fixed mean—rejected when the t-statistic on x_{t-1} falls below the 5% critical value of -2.86. The Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test of Kwiatkowski et al. (1992) reverses the null. It writes $x_t = \kappa_t + v_t$ with $\kappa_t = \kappa_{t-1} + u_t$ and tests $\mathrm{Var}\,(u_t) = 0$, the absence of the random-walk component κ_t, through the scaled partial sums of the demeaned series,

$$\hat{\eta} \; = \; \frac{1}{T^2} \frac{\sum\limits_{t=1}^{T} S_t^2}{\hat{s}^2(\ell)}, \qquad S_t \; = \; \sum_{i=1}^{t} (x_i - \bar{x})\,, \tag{A.1}$$

where $\hat{s}^2(\ell)$ is a Bartlett-weighted long-run variance of the same form as Eq. (A.3) below, computed on the demeaned series $x_t - \bar{x}$ rather than on the score and with its own bandwidth ℓ. Under stationarity the deviations $x_i - \bar{x}$ cancel and S_t stays of order $T^{1/2}$; under a unit root they accumulate and S_t grows like T, so $\hat{\eta}$ diverges. The test rejects for large values, above 0.463 at 5%.

A.2 HAC Inference for a Correlation

Writing the correlation of Eq. (8) as the slope of the ordinary least squares regression of the standardized benchmark on the standardized index, $\tilde{b}_t = \rho\,\tilde{E}_t + u_t$ with $\tilde{E}_t = \left(E_t - \bar{E}\right)/\hat{\sigma}_E$ and $\tilde{b}_t = \left(b_t - \bar{b}\right)/\hat{\sigma}_b$, the slope equals the Pearson correlation numerically. The estimator solves $T^{-1}\sum_{t=1}^{T} g_t\,(\hat{\rho}) = 0$ with score

$$g_t(\rho) = \tilde{E}_t \left(\tilde{b}_t - \rho\,\tilde{E}_t\right), \tag{A.2}$$

the contribution of day t to the least squares gradient. Generalized method of moments gives the sandwich variance $A^{-1}SA^{-1}/T$ with $A = \mathbb{E}\left[\tilde{E}_t^2\right]$ and $S = \sum_{j=-\infty}^{\infty} \mathrm{Cov}\,(g_t, g_{t-j})$ the long-run variance of the score. Standardizing the regressor sets $A = 1$, the sandwich collapses to S/T; this is the HAC variance of the standardized-slope estimand, which coincides numerically with $\hat{\rho}$ and conditions on the sample means and variances used to standardize. The full influence function of $\hat{\rho}$ adds the terms from estimating those moments; under the near-unit-root persistence documented in Section 4.2 the serial-dependence correction in S dominates them. We report the standardized-slope HAC standard error and, for reference, the classical Fisher standard error $\left(1 - \hat{\rho}^2\right)/\sqrt{T}$.

We estimate S with the heteroskedasticity- and autocorrelation-consistent (HAC) estimator of Newey and West (1987). Let $\hat{\gamma}_j = T^{-1}\sum_{t=j+1}^{T} \hat{g}_t\,\hat{g}_{t-j}$ denote the sample autocovariances of the fitted score. Truncating at lag L and weighting by the triangular window of Bartlett (1950) gives the long-run variance

$$\hat{s}^2(L) = \hat{\gamma}_0 + 2\sum_{j=1}^{L} w_j\,\hat{\gamma}_j, \qquad w_j = 1 - \frac{j}{L+1}, \tag{A.3}$$

and the variance of the estimated correlation is

$$\widehat{\mathrm{Var}}(\hat{\rho}) = \frac{\hat{s}^2(L)}{T}. \tag{A.4}$$

The weights are not cosmetic. An unweighted truncation ($w_j = 1$) can return a negative variance, whereas the Bartlett window is the autoconvolution of a rectangular window, its Fourier transform is the nonnegative Fejér kernel, and $\hat{s}^2(L) \geq 0$ holds by construction.

The bandwidth follows the autoregressive plug-in of Andrews (1991) applied to $\hat{g}_t$, which selects a long window for the VIX and a short one for RV_t. We report $L \in \{7, 30, 60, 100\}$, the lower end being the sample-size rule $L = \left\lfloor 4\,(T/100)^{2/9} \right\rfloor$ of Newey and West (1994), which equals 7 at $T = 1\,835$ and 6 at $T = 1\,142$.

The exponent 2/9 deserves a word, because it is not the rate at which the bandwidth itself should grow. For the Bartlett kernel the truncation point that minimises the mean squared error of $\hat{s}^2(L)$ grows at rate $T^{1/3}$, with a constant of proportionality that depends on the serial dependence of $\hat{g}_t$ and is therefore unknown. Newey and West (1994) estimate that constant from a preliminary, more crudely truncated sum of the sample autocovariances, and $\left\lfloor 4\,(T/100)^{2/9} \right\rfloor$ is the number of lags they allow that preliminary sum, not the bandwidth it delivers. The rate $T^{2/9}$ is deliberately slower than the $T^{1/3}$ of the bandwidth: a pilot that grew as fast as the object it is calibrating would carry enough variance of its own to contaminate the plug-in, while one that grows at $T^{2/9}$ still lets its bias vanish. The constant 4 and the normalisation by $T/100$ are their calibration for this kernel; the quadratic-spectral kernel takes $12\,(T/100)^{2/25}$ in the same role. What the rule is, then, is a function of the sample size alone: it knows nothing about how persistent our score actually is, which is why we report it as the low end of a grid rather than as the answer.

A.3 The GARCH$(1,1)$ Model

Let x_t denote the series being modelled and let $\mathcal{F}_{t-1} = \sigma(x_{t-1}, x_{t-2}, \ldots)$ denote the information set available at the end of day $t-1$; all conditional moments below are taken with respect to it. The conditional mean and conditional variance of the series are

$$m_t \;=\; \mathbb{E}[x_t \mid \mathcal{F}_{t-1}], \qquad \sigma_t^2 \;=\; \mathrm{Var}(x_t \mid \mathcal{F}_{t-1}) \;=\; \mathbb{E}\Big[(x_t - m_t)^2 \mid \mathcal{F}_{t-1}\Big], \tag{A.5}$$

the mean and dispersion of today's observation given everything observed through yesterday. The conditional variance is the object of interest: it is the day-by-day counterpart of the unconditional variance $\mathrm{Var}(x_t)$, and its square root σ_t, the conditional volatility, is the daily measure of turbulence the model delivers.

The recursion of Eq. (12) is initialized at σ_0^2 and, by induction, σ_t^2 is a measurable function of $x_{t-1}, \ldots, x_0$: it is known at $t-1$, as a conditional variance must be. Three further properties make the model interpretable.

First, taking unconditional expectations in Eq. (12) requires $\mathbb{E}[\varepsilon_{t-1}^2]$. Because σ_{t-1}^2 is $\mathcal{F}_{t-2}$-measurable and z_{t-1} is independent of $\mathcal{F}_{t-2}$ with $\mathbb{E}[z_{t-1}^2] = 1$, the law of iterated expectations gives

$$\mathbb{E}\Big[\varepsilon_{t-1}^2\Big] \;=\; \mathbb{E}\Big[\mathbb{E}\Big[\sigma_{t-1}^2\, z_{t-1}^2 \mid \mathcal{F}_{t-2}\Big]\Big] \;=\; \mathbb{E}\Big[\sigma_{t-1}^2\, \mathbb{E}\Big[z_{t-1}^2 \mid \mathcal{F}_{t-2}\Big]\Big] \;=\; \mathbb{E}\Big[\sigma_{t-1}^2\Big].$$

In stationarity, $\mathbb{E}[\sigma_t^2] = \mathbb{E}[\sigma_{t-1}^2] = \bar{\sigma}^2$, so $\bar{\sigma}^2 = \omega + (\alpha+\beta)\,\bar{\sigma}^2$, that is,

$$\bar{\sigma}^2 \;=\; \frac{\omega}{1-\alpha-\beta}, \tag{A.6}$$

which is why $\omega > 0$ and $\alpha + \beta < 1$ are imposed. Second, iterating the conditional expectation of Eq. (12) forward gives the h-step variance forecast

$$\mathbb{E}\Big[\sigma_{t+h}^2 \mid \mathcal{F}_t\Big] - \bar{\sigma}^2 \;=\; (\alpha+\beta)^{\,h-1}\left(\sigma_{t+1}^2 - \bar{\sigma}^2\right), \qquad h \geq 1,$$

so deviations of volatility from its long-run level die out geometrically at rate $\alpha+\beta$, and a volatility shock has an implied half-life of $\ln(1/2)/\ln(\alpha+\beta)$ days.

Third, repeated back-substitution of $\beta\sigma_{t-1}^2$ yields

$$\sigma_t^2 \;=\; \frac{\omega}{1-\beta} + \alpha\sum_{j=0}^{\infty}\beta^{\,j}\,\varepsilon_{t-1-j}^2,$$

an ARCH model of infinite order with geometrically decaying weights: the single pair (α, β) encodes a complete memory profile, which is what makes the $(1,1)$ order sufficient in practice.

Estimation is by Gaussian maximum likelihood. The distributional assumption implies $\varepsilon_t \mid \mathcal{F}_{t-1} \sim \mathrm{N}(0, \sigma_t^2)$, so the joint density factorizes by the prediction-error decomposition into the product of these one-step conditional densities, $f(x_1, \ldots, x_T; \theta) = \prod_t f(x_t \mid \mathcal{F}_{t-1}; \theta)$ with $\theta = (\mu, \phi, \omega, \alpha, \beta)$, and the log-likelihood is

$$\ell(\theta) \;=\; -\frac{1}{2}\sum_{t=1}^{T}\left[\ln(2\pi) + \ln\sigma_t^2(\theta) + \frac{\varepsilon_t(\theta)^2}{\sigma_t^2(\theta)}\right], \tag{A.7}$$

where $\varepsilon_t(\theta)$ and $\sigma_t^2(\theta)$ are computed recursively from Eqs. (11)–(12), the recursion being initialized at the sample variance of the fitted innovations. The estimator $\hat{\theta}$ maximizes $\ell(\theta)$; no closed form exists, and the maximization is carried out numerically from several starting points.

A.4 The Area Under the ROC Curve

Section 6.1 scores each series by how well it separates the crisis dates of Table 7 from the calm ones. This appendix states what that statistic is, why it needs no model and no threshold, and where its standard error comes from.

The curve. Fix a series X_t and read it as a classifier: declare day t a crisis whenever $X_t > c$. Let K denote the set of n_k crisis dates and C the set of n_c calm dates. Each threshold c produces one true positive rate and one false positive rate,

$$\mathrm{TPR}(c) = \Pr\left(X > c \mid K\right), \qquad \mathrm{FPR}(c) = \Pr\left(X > c \mid C\right), \tag{A.8}$$

and the receiver operating characteristic curve is the locus $\{(\mathrm{FPR}(c), \mathrm{TPR}(c)) : c \in \mathbb{R}\}$ traced as c moves across the support of X. Both rates fall from one to zero as c rises, so the curve runs from $(1,1)$ to $(0,0)$. A series carrying no information about the split gives $\mathrm{TPR}(c) = \mathrm{FPR}(c)$ at every c and the curve is the diagonal. The area under it,

$$\mathrm{AUC} = \int_0^1 \mathrm{TPR}\left(\mathrm{FPR}^{-1}(u)\right)\mathrm{d}u, \tag{A.9}$$

summarizes the whole family of thresholds in one number and is what Table 9 reports.

What it estimates. Let X^K be the value of the series on a randomly drawn crisis date and X^C its value on a randomly drawn calm date, the two draws independent. Then

$$\mathrm{AUC} = \Pr\left(X^K > X^C\right) + \tfrac{1}{2}\Pr\left(X^K = X^C\right), \tag{A.10}$$

the probability that the series ranks a crisis day above a calm day, with ties split. The area is therefore a statement about ranking, not about levels, and it is estimated by the two-sample U-statistic of Mann and Whitney (1947),

$$\widehat{\mathrm{AUC}} = \frac{1}{n_k n_c}\sum_{t\in K}\sum_{s\in C}\psi\left(X_t, X_s\right), \qquad \psi(a,b) = \mathbf{1}\{a>b\} + \tfrac{1}{2}\mathbf{1}\{a=b\}, \tag{A.11}$$

which is the Mann–Whitney statistic divided by $n_k n_c$. Three properties follow and are the reason we use it.

First, $\widehat{\mathrm{AUC}}$ depends on the data only through the ranks of the pooled sample, so it is invariant to any strictly increasing transformation of X_t. The base-100 rescaling of Eq. (5) therefore cannot affect it, and neither could a logarithm or any monotone reweighting; nothing in Table 9 turns on the units of the index. Second, the estimand conditions on the two populations separately, so it does not depend on how many crisis dates the calendar contains relative to calm ones. A window that is short is not thereby penalised, which matters here because the eight windows range from seven to seventy trading days. Third, reversing the sign of the series reverses the area,

$$\mathrm{AUC}\left(-X\right) = 1 - \mathrm{AUC}\left(X\right), \tag{A.12}$$

so $\frac{1}{2}$ is the uninformative value and the discriminating content of a series is $\left|\mathrm{AUC} - \frac{1}{2}\right|$. A value below one half is not a failed detector but an inverted one, carrying exactly as much information as its mirror image. This is what licenses the reading of Section 6.1 under which the headline index at 0.327 on the political windows is as informative about them as it is at 0.939 on the economic ones, with the sign telling us which way the discourse moves.

Its sampling error. Because Eq. (A.11) is a two-sample U-statistic of degree $(1,1)$, its variance follows from the Hájek projection. Define the placement values

$$V_t = \frac{1}{n_c}\sum_{s\in C}\psi\left(X_t, X_s\right),\ t\in K, \qquad W_s = \frac{1}{n_k}\sum_{t\in K}\psi\left(X_t, X_s\right),\ s\in C, \tag{A.13}$$

the fraction of calm days that a given crisis day outranks, and the fraction of crisis days that a given calm day is outranked by. Both average to $\widehat{\mathrm{AUC}}$. Writing $S_{10} = \mathrm{Var}(V_t)$ and $S_{01} = \mathrm{Var}(W_s)$, DeLong et al. (1988) give

$$\mathrm{Var}\left(\widehat{\mathrm{AUC}}\right) = \frac{S_{10}}{n_k} + \frac{S_{01}}{n_c}. \tag{A.14}$$

For two series scored on the same pair of samples, the placement values are formed for both and S_{10} and S_{01} become 2×2 covariance matrices across series; the variance of the difference of the two areas is then $L'\left(S_{10}/n_k + S_{01}/n_c\right)L$ with $L = (1,-1)'$. This is the statistic reported in Panel C of Table 9.

Eq. (A.14) divides the dispersion of the placement values by the sample sizes, which is the variance of a mean of independent draws. Our dates are not independent: the index is a trailing fourteen-day rate, so consecutive placement values are nearly identical and the effective number of observations is far below n_k and n_c. The DeLong standard error is therefore too small here, and the interval we rely on is the circular block bootstrap described in the note to Table 9, which resamples the joint time series in blocks and so preserves that dependence. The z statistic is reported for reference only.

Table 1. Complete List of X Accounts (97)

Username	Name	Category
RosanaCuevaM	Rosana Cueva	Journalist
NicolasLucar	Nicolás Lucar	Journalist
PolloFarsantePe	Beto Ortiz	Journalist
MilagrosLeivaG	Milagros Leiva	Journalist
DeltaMdelta	Mónica Delta	Journalist
MavilaHuertasC	Mavila Huertas	Journalist
SolCn	Sol Carreño	Journalist
VertizPamela	Pamela Vértiz	Journalist
claudia_cisneros	Claudia Cisneros	Journalist
recisneros	Renato Cisneros	Journalist
larryportera	Paola Ugaz	Journalist
tuesta	Fernando Tuesta Soldevilla	Journalist
Federicoagust	Federico Salazar B.	Journalist
julianaoxenford	Juliana Oxenford	Journalist
mariateguiperu	Aldo Mariátegui	Journalist
JaimeChincha	Jaime Chincha	Journalist
VeroLinaresC	Verónica Linares	Journalist
tenoriopedro	Pedro Tenorio	Journalist
JBCPERU	José Barba Caballero	Journalist
jdealthaus	Jaime de Althaus	Journalist
ensustrece	Hildebrandt en sus trece	Journalist
BaylyOficial	Jaime Bayly	Journalist
jctafur	Juan Carlos Tafur	Journalist
cparedesr	Carlos Paredes	Journalist
PCaterianoB	Pedro Cateriano B.	Politician
Carlos_Bruce	Carlos Bruce	Politician
JuanSheput	Juan Sheput	Politician
VLADIMIR_CERRON	Vladimír Cerrón	Politician
anibaltorresv	Aníbal Torres V.	Politician
pedrofrancke	Pedro Francke	Politician
Ollanta_HumalaT	Ollanta Humala Tasso	Politician
sigridbazan	Sigrid Bazán	Politician
MirtyVas	Mirtha Vásquez	Politician
FlorPabloMedina	Flor Pablo Medina	Politician
rlopezaliaga1	Rafael López Aliaga	Politician
patarevalo	Patricia Arévalo	Politician
George_Forsyth	George Forsyth	Politician
JorgeMunozPe	Jorge Muñoz	Politician
DanielUrresti1	Daniel Urresti	Politician
VictorAndresGB	Víctor Andrés García Belaunde	Politician
KeikoFujimori	Keiko Fujimori	Politician
MartinVizcarraC	Martín Vizcarra	Politician
JorgeDCG	Jorge del Castillo	Politician
Mauriciomulder	Mauricio Mulder	Politician

Username	Name	Category
nidiavilchez	Nidia Vílchez	Politician
FSagasti	Francisco Sagasti	Politician
AlbertoBelaunde	Alberto de Belaunde	Politician
MaricarmenAlvaP	Maricarmen Alva	Politician
PattyChirinos1	Patricia Chirinos	Politician
adrianatudelag	Adriana Tudela Gutiérrez	Politician
CesarAcunaP	César Acuña Peralta	Politician
MesiasGuevara	Mesías Guevara	Politician
vozdelatierra	Marco Arana	Politician
MarisaGlave	Marisa Glave	Politician
GuilleBermejoR	Guillermo Bermejo Rojas	Politician
robertochiabra	Roberto Chiabra	Politician
RichardArcePeru	Richard Arce	Politician
Alm_Montoya	Jorge Montoya	Politician
AlejandroCavero	Alejandro Cavero	Politician
HectorValer_PER	Héctor Valer Pinto	Politician
RosselliAmuruz	Rosselli Amuruz	Politician
HDeSotoPeru	Hernando de Soto	Politician
UrsulaLetonaP	Ursula Letona	Politician
AlvaroVargasLl	Álvaro Vargas Llosa	Politician
AlbertoOtarolaP	Alberto Otárola	Politician
HPerezDeCuellar	Hania Pérez de Cuéllar	Politician
alosegura	Alonso Segura Vasi	Economist
BustamantePao18	Paola Bustamante S.	Economist
aethorne	Alfredo E. Thorne	Economist
WaldoMendozaB	Waldo Mendoza Bellido	Economist
JuanManuelGarc3	Juan Manuel García C.	Economist
CarlosAnderso1	Carlos A. Anderson	Economist
hugonopo	Hugo Ñopo	Economist
tuestadavid	David Tuesta	Economist
pablo_lavado	Pablo Lavado	Economist
OswaldoMolinaC	Oswaldo Molina	Economist
CParodiT	Carlos Parodi Trece	Economist
psecadae	Pablo Secada	Economist
DWinkelried	Diego Winkelried	Economist
romerocaroperu	Manuel Romero Caro	Economist
manupulgarvidal	Manuel Pulgar Vidal	Economist
BancalariA	Antonella Bancalari	Economist
eljorobado	Carlos Meléndez	Political analyst
DargentEduardo	Eduardo Dargent	Political analyst
DelaPuenteJuan	Juan De la Puente	Political analyst
lauermirko	Mirko Lauer	Political analyst
rmapalacios	Rosa María Palacios	Political analyst
Frospigliosi	Fernando Rospigliosi	Political analyst
BetoAdrianzen	Alberto Adrianzén	Political analyst
ocram	Marco Sifuentes	Political analyst

Username	Name	Category
brunoschaaf	Bruno Schaaf	Political analyst
luisbenaventegi	Luis Benavente	Political analyst
camcesar	César Campos	Political analyst
AlfredoMTorres	Alfredo M. Torres G.	Political analyst
gonza_banda	Gonzalo Banda	Political analyst
Roque_Benavides	Roque Benavides	Private sector
CayetanaAljovin	Cayetana Aljovín	Private sector

Table 2. Label Combinations (E, P, U).

E	P	U	Interpretation	Count
0	0	0	Unrelated to economics, politics, or uncertainty	81 353
1	0	0	Economic content without politics or uncertainty	2 051
0	1	0	Political content without economics or uncertainty	19 280
0	0	1	Uncertainty not linked to economics or politics	23 144
1	1	0	Economic and political content, no uncertainty	9 674
1	0	1	Economic uncertainty	1 318
0	1	1	Political uncertainty	59 978
1	1	1	Economic and political uncertainty (strict EPU)	16 451

Note: the eight cells partition the 213 249 classified tweets and sum to that total.

Table 3. Descriptive Statistics.

Statistic	Value
Total tweets	213 249
Unique accounts	97
Date range	2018-01-23 to 2023-01-31
Calendar days	1 835
Daily N_t: Mean / Median / Max	116.21 / 56 / 1 016
Total strict EPU tweets ($\sum_t Y_t$)	16 451
Days with $Y_t > 0$	1 462
Y_t mean (cond. on $Y_t > 0$)	11.25
Y_t maximum	82

Note: N_t is the daily tweet volume and Y_t the daily strict-EPU count ($E \cap P \cap U$); the conditional mean is taken over the 1 462 days with $Y_t > 0$.

Table 4. Persistence and Stationarity Diagnostics.

Series	ADF t	Lags	KPSS $\hat{\eta}$	$\hat{\rho}_1$	Half-life (days)
EPU_t	-3.84	11	1.726	0.967	20.6
VIX_t	-3.88	4	3.775	0.964	18.8
$\mathrm{RV}_t = r_t^2$	-12.27	3	1.086	0.168	0.4
$\hat{\sigma}_{\mathrm{BVL},t}$	-2.92	1	5.400	0.987	54.7

Note: $T = 1\,142$ observations on the common trading-day panel of Section 4.1, 24 January 2018 to 31 January 2023. Like every other financial statistic in the paper, these diagnostics are computed on the trading-day panel and not on the interpolated calendar panel; the previous version reported them on the latter, where the fill inflates the measured persistence of every market series. Augmented Dickey–Fuller regression with a constant and no trend; the lag order is chosen by the Bayesian information criterion over $p \leq \left\lfloor 12\,(T/100)^{1/4} \right\rfloor = 22$, and the 5% critical value is -2.86. The KPSS statistic of Eq. (A.1) tests level stationarity with a Bartlett bandwidth $\ell = \left\lfloor 4\,(T/100)^{1/4} \right\rfloor = 7$; the 5% critical value is 0.463. $\hat{\rho}_1$ is the first-order sample autocorrelation and the half-life is $\ln(1/2)/\ln \hat{\rho}_1$. All four series reject the unit root and also reject stationarity, the signature of the near-unit-root region, except RV_t, which reverts within a day; its KPSS rejection reflects the rise in its mean from 4.4×10^{-5} before 2020 to 2.6×10^{-4} afterwards rather than serial dependence.

Table 5. GARCH(1, 1) Estimates.

	r_t (BVL)		VIX_t		EPU_t	
Innovation	Normal	Student-t	Normal	Student-t	Normal	Student-t
Mean equation	Constant	Constant	AR(1)	AR(1)	AR(1)	AR(1)
$\hat{\phi}$	—	—	0.980	0.971	0.974	0.983
$\hat{\omega}$	0.0069	0.0231	6.877	2.869	3.346	1.990
$\hat{\alpha}$	0.0602	0.0704	0.4610	0.2741	0.0849	0.1090
$\hat{\beta}$	0.9397	0.9216	0.5361	0.7258	0.8859	0.8909
$\hat{\nu}$	—	3.60	—	3.26	—	3.70
Persistence $\hat{\alpha} + \hat{\beta}$	$0.9999^{\dagger}$	0.9920	0.9972	$0.9999^{\dagger}$	0.9708	$0.9999^{\dagger}$
Volatility half-life (days)	—	85.8	243.9	—	23.4	—
Filter memory $1/(1 - \hat{\beta})$	16.6	12.8	2.2	3.6	8.8	9.2
Log-likelihood	$-1\,763.3$	$-1\,666.5$	$-3\,756.3$	$-3\,576.9$	$-4\,256.4$	$-4\,192.7$
Observations	1 142	1 142	1 141	1 141	1 141	1 141
$\hat{\rho}_1(\hat{z}_t^2)$	0.046	0.037	-0.008	0.001	0.015	0.012
Ljung–Box(10) p-value	0.75	0.89	1.00	1.00	0.24	0.45
Memorandum: BVL return on the interpolated calendar panel, the specification of the previous version						
$\hat{\alpha}$ / $\hat{\beta}$ / persistence	0.0799 / 0.9180 / 0.9979; half-life 332.2 days; $T = 1\,834$; Ljung–Box $p = 0.06$					

Note: GARCH(1, 1) of Eqs. (11)–(12), estimated by maximum likelihood written from scratch and maximised from ten starting points. The filter is applied to the BVL log-return r_t, never to r_t^2: r_t^2 is already a variance proxy, and a GARCH fitted to it would model the variance of a variance. The Student-t columns are the robustness check: the innovation is a standardised t with $\hat{\nu}$ degrees of freedom, estimated jointly. All columns use the common trading-day panel of Section 4.1. $\hat{\nu}$ near 3.5 in all three series says the standardised innovation has no fourth moment, which is why the Gaussian fit is a quasi-likelihood. A dagger marks a persistence estimate at the numerical upper bound of 0.9999: the likelihood is maximised at the IGARCH boundary, the half-life is not identified, and we report no value for it rather than a spurious one. The memorandum line reproduces the calendar-panel estimate of the previous version and is retained so the contrast is visible: interpolating the non-trading dates lowers $\hat{\alpha}$, raises $\hat{\beta}$ and drives persistence to 0.9979 with an implied half-life of 332 days, an artifact of the fill rather than a property of the Lima market.

Table 6. Comparison of the Index and the Two Benchmarks.

	EPU index	VIX	BVL volatility
Underlying data	Elite tweets	Index options	Squared returns
Geographic origin	Peru	United States	Peru
Uncertainty type	Political, economic	Financial (implied)	Financial (realized)
Temporal orientation	Contemporaneous	Forward-looking	Backward-looking
Price-anchored	No	Yes	Yes
Availability	Calendar days	Trading days	Trading days

Note: the VIX is option-implied from Standard & Poor's 500 (S&P 500) options and is forward-looking over about one month; BVL volatility is realized from squared log-returns of the Bolsa de Valores de Lima index and, in the GARCH$(1,1)$ analysis, filtered as a conditional volatility; the index reflects uncertainty as it is expressed in discourse. Price-anchored indicates whether the measure derives from a traded price.

Table 7. Crisis Windows in Peru, 2018–2023.

ID	Start	End	Days	Episode
C1	2018-03-12	2018-04-06	26	Fall of Kuczynski; transition to Vizcarra
C2	2019-09-27	2019-10-15	19	Dissolution of Congress
C3	2020-03-15	2020-06-30	108	COVID-19 emergency and lockdown
C4	2020-11-09	2020-11-18	10	Vizcarra vacancy; Merino interregnum
C5	2021-04-08	2021-07-28	112	Contested general election
C6	2022-03-28	2022-04-12	16	Fuel-price protests; Lima curfew
C7	2022-12-07	2022-12-20	14	Castillo self-coup and removal
C8	2022-12-20	2023-01-31	43	Post-Castillo protests
Union of windows			347	18.9% of the 1 835 sample days

Note: windows are dated from the historical record, not market volatility, keeping the crisis/calm split exogenous to the benchmarks. Inclusive calendar days. C7 and C8 share 2022-12-20, counted once in the union ($26 + 19 + 108 + 10 + 112 + 16 + 14 + 43 = 348$ window days; union = 347). C8 is right-censored by the sample end. Edges flagged as judgment calls are checked by shifting all windows ±5 days.

Table 8. Static Co-Movement of the Index with the Two Benchmarks.

Sample	T	$\hat{\rho}$	$\hat{\rho}_S$	Standard error of $\hat{\rho}$: Classical	$L=7$	$L=30$	$L=60$	$L=100$	L^A
Panel A. Benchmark $b_t = \mathrm{VIX}_t$									
Full	1 142	0.429	0.426	0.024	0.069	0.082	0.080	0.075	0.076
Crisis	218	0.581	0.607	0.045	0.102	0.074	0.062	0.055	0.100
Calm	924	0.315	0.375	0.030	0.067	0.087	0.091	0.094	0.069
Panel B. Benchmark $b_t = r_t^2$									
Full	1 142	0.061	0.122	0.029	0.039	0.033	0.032	0.033	0.038
Crisis	218	0.021	0.024	0.068	0.073	0.074	0.078	0.066	0.069
Calm	924	0.056	0.149	0.033	0.044	0.049	0.054	0.057	0.042

Note: $\hat{\rho}$ is the Pearson correlation of Eq. (8) between the headline index EPU_t and the benchmark b_t, and $\hat{\rho}_S$ the Spearman correlation of Eq. (9), both computed on the common trading-day panel of Section 4.1 rather than on the interpolated calendar panel. Classical is $\left(1-\hat{\rho}^2\right)/T^{1/2}$. The remaining columns are the Newey–West standard error of Eq. (A.4) at the stated Bartlett bandwidth; L^A is the Andrews autoregressive plug-in bandwidth, equal to 10, 6 and 8 for the VIX rows and to 5, 4 and 4 for the r_t^2 rows. These bandwidths are an order of magnitude smaller than the 99, 39 and 55 of the previous version, which is the expected consequence of removing the interpolated dates: the fill was inducing most of the serial dependence the plug-in was responding to. Crisis dates are the union of the eight windows of Table 7 restricted to trading dates.

Table 9. Crisis Discrimination: Area Under the Receiver Operating Characteristic Curve.

ID	Type	Days	EPU_t	PU_t	VIX_t	r
Panel A. Window by window, against the 924 calm trading days						
C1	Political	18	0.607	0.947	0.437	0.43
C2	Political	12	0.380	0.874	0.363	0.53
C3	Economic	66	0.962	0.675	0.919	0.64
C4	Political	8	0.101	0.611	0.695	0.74
C5	Political	70	0.289	0.497	0.483	0.63
C6	Economic	12	0.813	0.872	0.586	0.43
C7	Political	7	0.392	0.999	0.686	0.58
C8	Political	26	0.257	0.927	0.598	0.52
Panel B. Pooled by episode type, adding the GARCH volatility as a fourth series						
			EPU_t	PU_t	VIX_t	r
All windows		218	0.546	0.699	0.638	0.59
Economic (C3, C6)		78	0.939	0.705	0.868	0.61
Political (C1–C2, C4–C5, C7–C8)		140	0.327	0.695	0.509	0.58
Political, excluding C5		70	0.364	0.893	0.535	0.53
$\hat{\sigma}_{\text{BVL},t}$ (GARCH on r_t)			0.803 (all), 0.917 (economic), 0.739 (political), 0.560 (political ex-C5			

Panel C. Paired difference $\widehat{\text{AUC}}\,(\text{PU}_t) - \widehat{\text{AUC}}\,(\text{EPU}_t)$

Subsample	Days	Difference	DeLong z	Block-bootstrap 95% interval
All windows	218	+0.153	5.50	[−0.044, +0.348]
Economic	78	−0.234	−9.55	[−0.350, −0.097]
Political	140	+0.368	14.06	[+0.213, +0.524]
Political, excluding C5	70	+0.529	16.69	[+0.340, +0.689]

Note: entries in Panels A and B are $\widehat{\text{AUC}}$, the Mann–Whitney estimate of $\Pr(X_t > X_s)$ for a crisis date t and a calm date s, computed on the common trading-day panel of Section 4.1. The kernel weights ties at one half, $\psi(a,b) = \mathbf{1}\{a > b\} + \frac{1}{2}\mathbf{1}\{a = b\}$; the previous version used the strict inequality alone, which biases the statistic downward whenever the series repeats a value. Crisis dates are the windows of Table 7, dated from the historical record and therefore exogenous to every series compared here. Panel A widths sum to 219 rather than the 218 of Panel B because C7 and C8 share 20 December 2022. A value of 0.5 is the coin flip and a value below it means the series runs lower inside the window.

Panel C reports the difference between two areas estimated on the same pair of samples. The DeLong z is computed from the full DeLong et al. (1988) covariance: the placement values $V_t = n_c^{-1}\sum_s \psi(X_t, X_s)$ and $W_s = n_k^{-1}\sum_t \psi(X_t, X_s)$ of Eq. (A.13) are formed for *both* series, and the variance of the difference is $L'\left(S_{10}/n_k + S_{01}/n_c\right)L$ with $L = (1,-1)'$ and S_{10}, S_{01} the covariance matrices across series; Appendix A.4 derives this. This is the correction of the previous version, which reported a z inconsistent with its own bootstrap interval.

The z is nevertheless reported for reference only and no inference rests on it. The DeLong variance divides the dispersion of the placement values by n_k and n_c, which is the variance of a mean of independent draws; the index is a trailing fourteen-day rate, so consecutive placement values are nearly identical and the effective sample is far smaller than the nominal one. The valid interval is the circular block bootstrap, mean block length 20 trading days and 2 000 replications, which resamples the joint time series and so preserves that dependence. Its standard errors exceed the DeLong ones by factors of 2.7 to 3.6 across the four subsamples, close to the $\sqrt{14} \approx 3.7$ that a fourteen-day dependence block implies. Note that the pooled row over all windows, significant under DeLong at $z = 5.50$, has a bootstrap interval that contains zero.

Table 10. Three Uncertainty Measures on a Common Scale.

	EPU_t	VIX_t	r_t^2	$\hat{\sigma}_{\mathrm{BVL},t}$
EPU_t	1			
VIX_t	0.4292	1		
r_t^2	0.0611	0.3170	1	
$\hat{\sigma}_{\mathrm{BVL},t}$	0.4529	0.6837	0.2183	1
Correlation with EPU_t *by regime, with HAC standard error*				
Full sample	—	0.429 (0.076)	0.061 (0.038)	0.453 (0.097)
Crisis	—	0.582 (0.101)	0.021 (0.069)	0.468 (0.139)
Calm	—	0.315 (0.069)	0.056 (0.042)	0.424 (0.050)

Note: Pearson correlations on the common trading-day panel, $T = 1\,142$. The four series are the headline index, the VIX, the one-day realised variance r_t^2, and the conditional volatility $\hat{\sigma}_{\mathrm{BVL},t}$ from the Gaussian GARCH$(1,1)$ fitted to r_t in Table 5. The Student-t filter delivers a volatility series correlated at 0.993 with the Gaussian one, so nothing in this table turns on the innovation assumption. Standard errors in parentheses are Newey–West with the Andrews plug-in bandwidth. The ordering is the substantive result: the two market volatilities agree most (0.684), which confirms the filter recovers a genuine volatility; the index sits alongside both at 0.43 and 0.45; and r_t^2 is nearly orthogonal to the index (0.061, 1.6 HAC standard errors from zero), because a squared daily return reverts within a day while the index averages a fortnight. Filtering is what makes the Lima market comparable to a fortnightly text-based index, which is why the GARCH step is carried out at all.

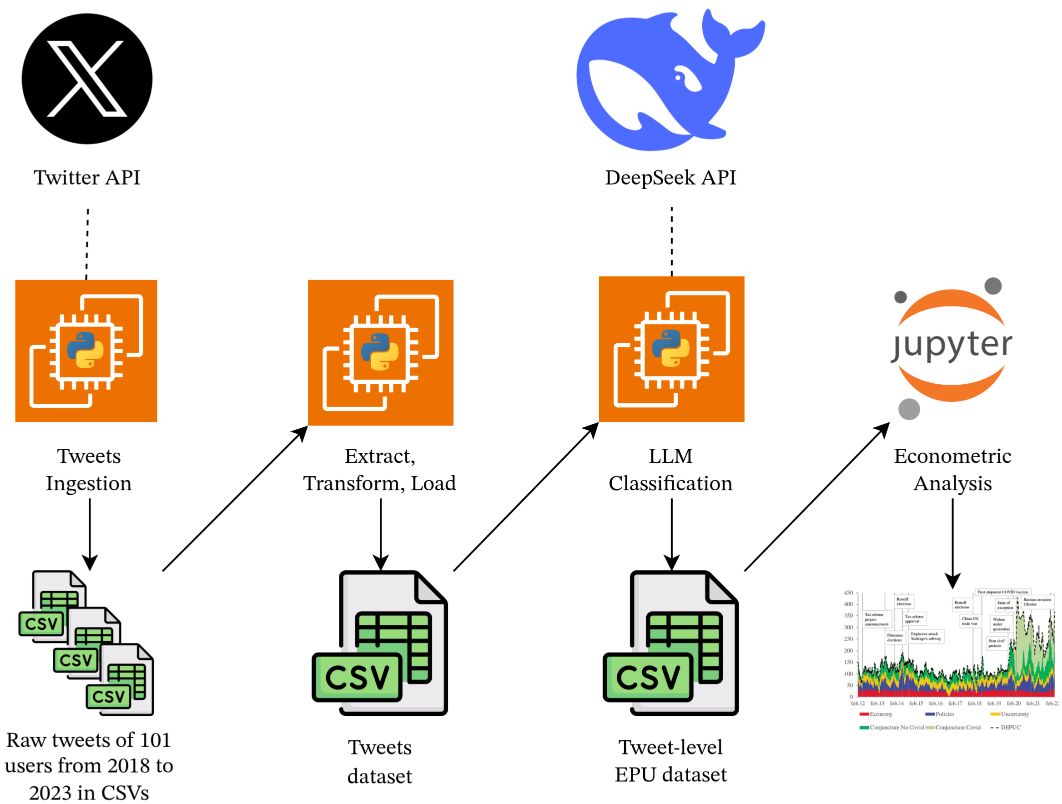

Figure 1. Pipeline Architecture, from Tweet Ingestion to Econometric Analysis. The final sample comprises the 97 accounts selected by the criteria of Section 2; any larger initial count shown in the schematic is illustrative of the pre-selection stage.

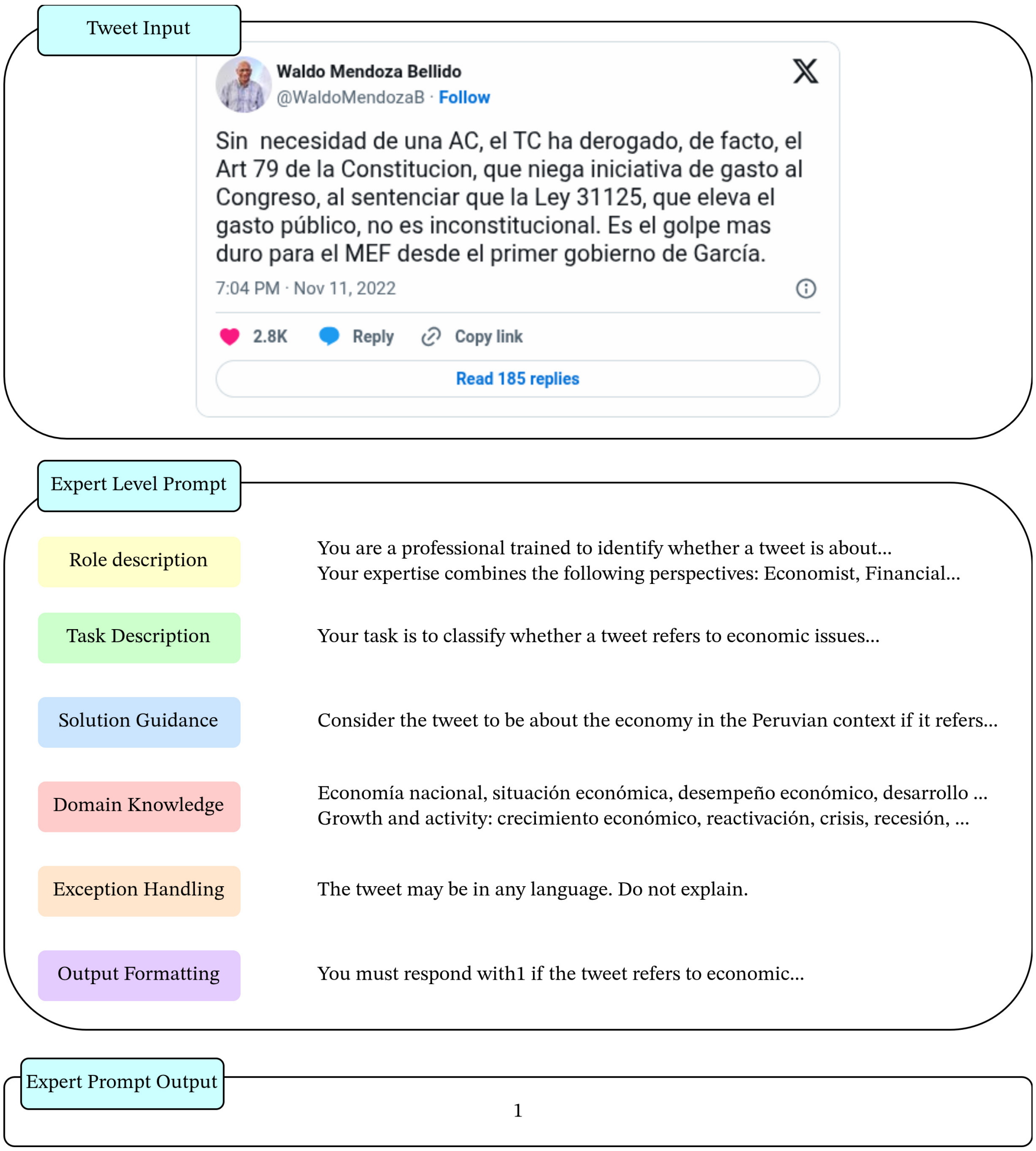

Figure 2. Expert-Level Prompt Structure for LLM Classification.

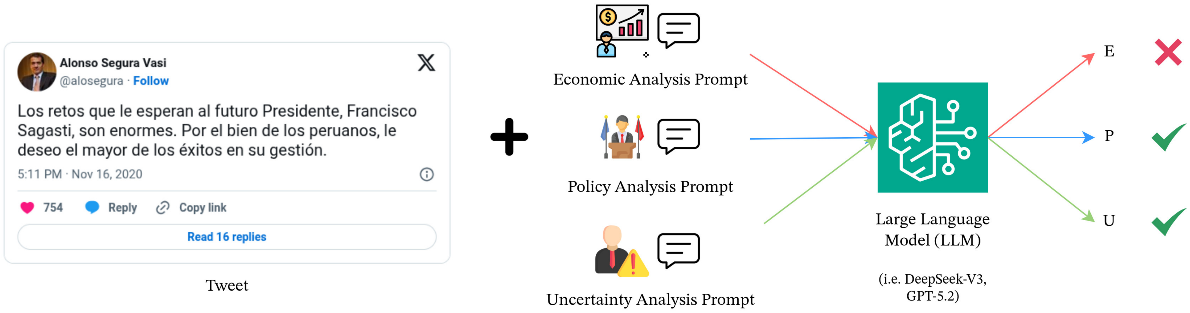

Figure 3. Three-Dimensional Labelling: Each Tweet Is Classified Independently for E_{ijt}, P_{ijt}, U_{ijt}.

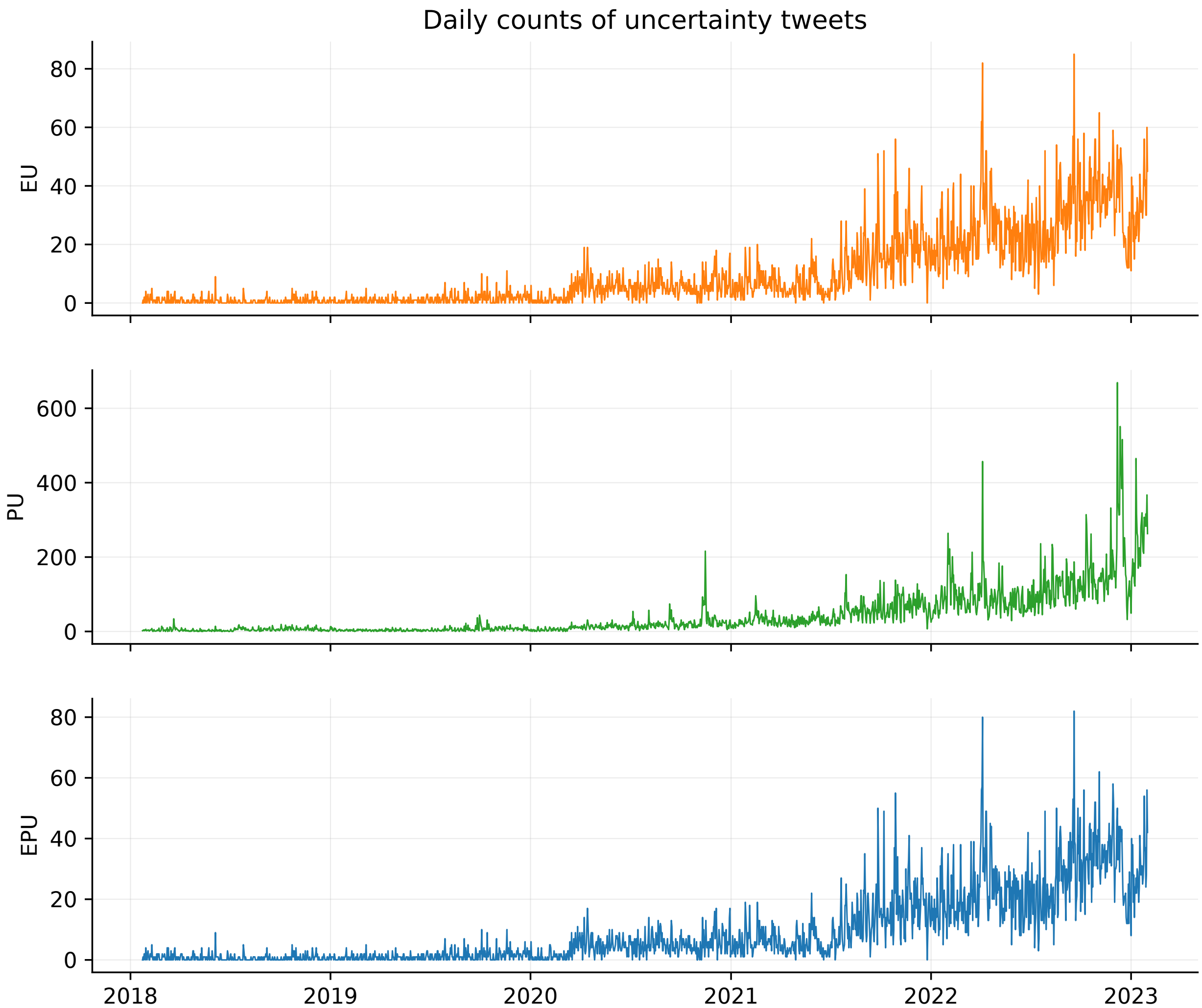

Figure 4. Daily Counts by Cell.

Daily counts of tweets that are, top to bottom, economic and uncertain ($E \cap U$), political and uncertain ($P \cap U$), and economic, political, and uncertain at once ($E \cap P \cap U$, the strict cell of Eq. (1)). The strict count and $E \cap U$ are nearly identical; $P \cap U$ is an order of magnitude larger and differently timed. All three rise with posting volume. Sample January 2018–January 2023; 97 accounts.

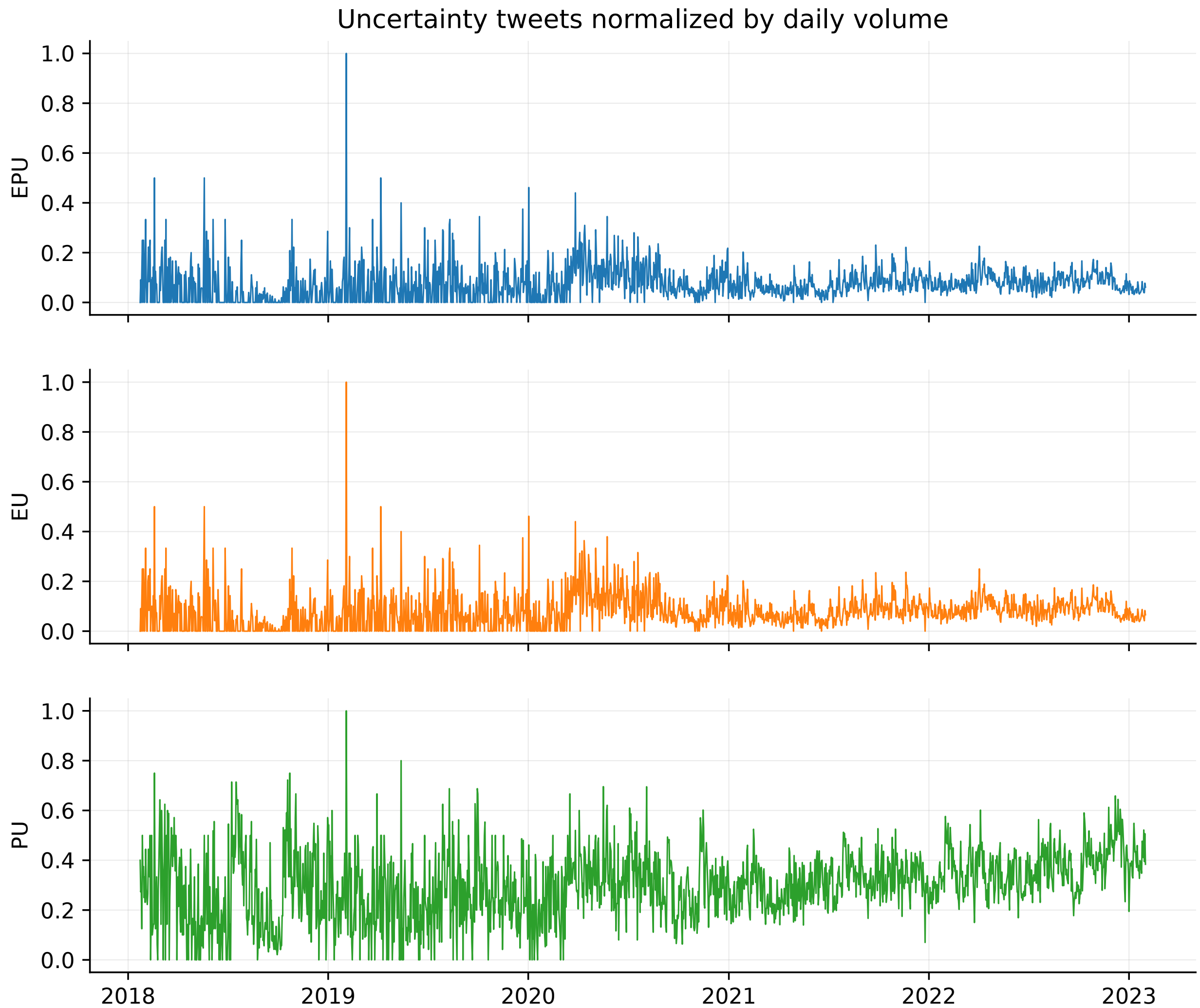

Figure 5. Daily Shares.

The daily share $s_t = Y_t/N_t$ (Eq. (2)) for the strict cell and the two broader cells. Normalizing by daily volume removes the upward drift of Figure 4 but leaves heavy small-sample noise on low-volume days, when a tiny denominator lets a single tweet move the share sharply.

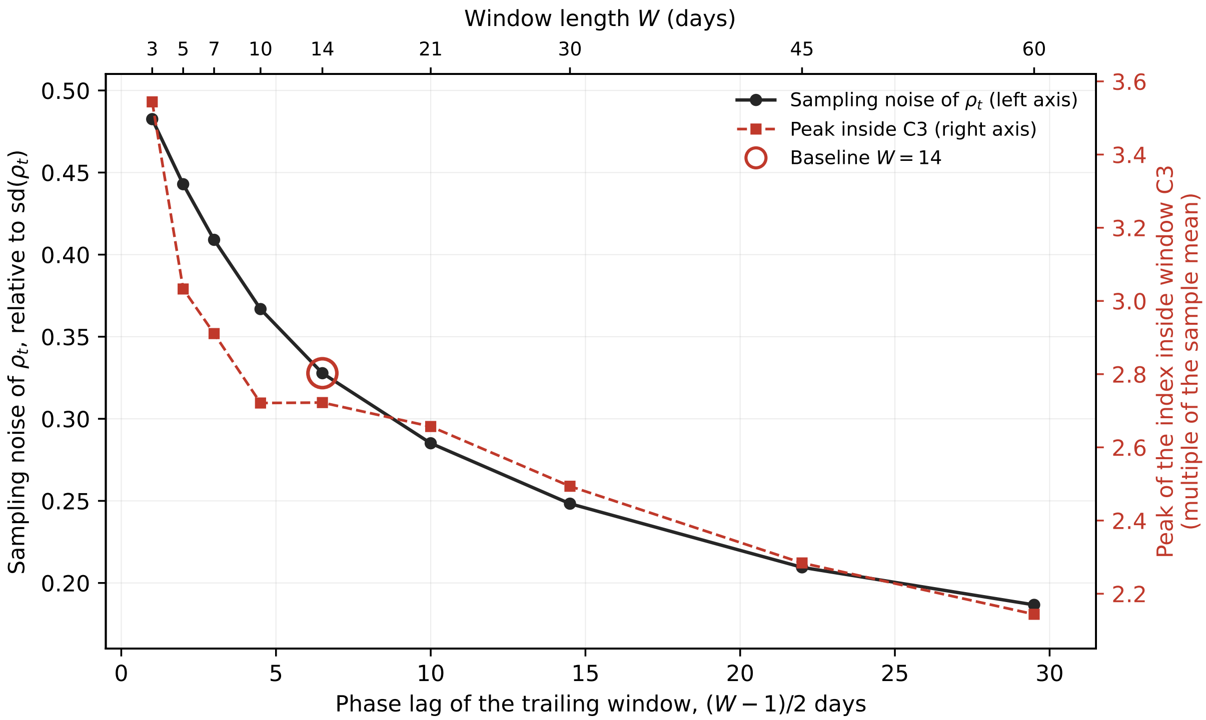

Figure 6. The Window-Length Trade-Off.

Each point is one trailing window W of Eq. (3), read on the top axis and placed on the bottom axis at the phase lag $(W-1)/2$ it imposes. Black, left axis: the mean sampling error of ρ_t, $(\rho_t(1-\rho_t)/\mathcal{N}_t)^{1/2}$, divided by the standard deviation of ρ_t, a scale-free noise ratio rather than a formal variance share. Red, right axis: the peak the index reaches inside the pandemic window C3, on 15 April 2020, in multiples of the sample mean. Pooling more days buys precision and pays in lag and amplitude; both curves fall monotonically, so no W minimizes them jointly. The circled baseline $W = 14$ costs a 6.5-day lag, a noise ratio of 0.33, and a retained peak of 2.72.

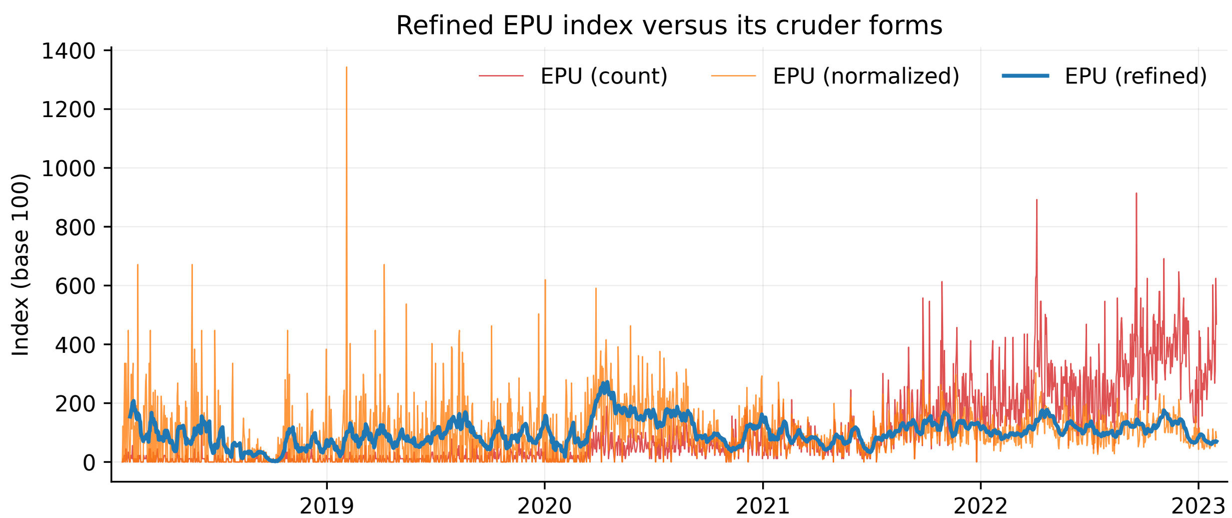

Figure 7. Refined Index Versus Simplier Forms.

The headline index (refined: the 14-day volume-weighted rate of Eq. (3), scaled to base 100 by Eq. (5)) against its raw count and normalized-share counterparts on the same scale. The simpler forms swing to many times their average; the refined index stays near its base and retains the episodes. Base 100.

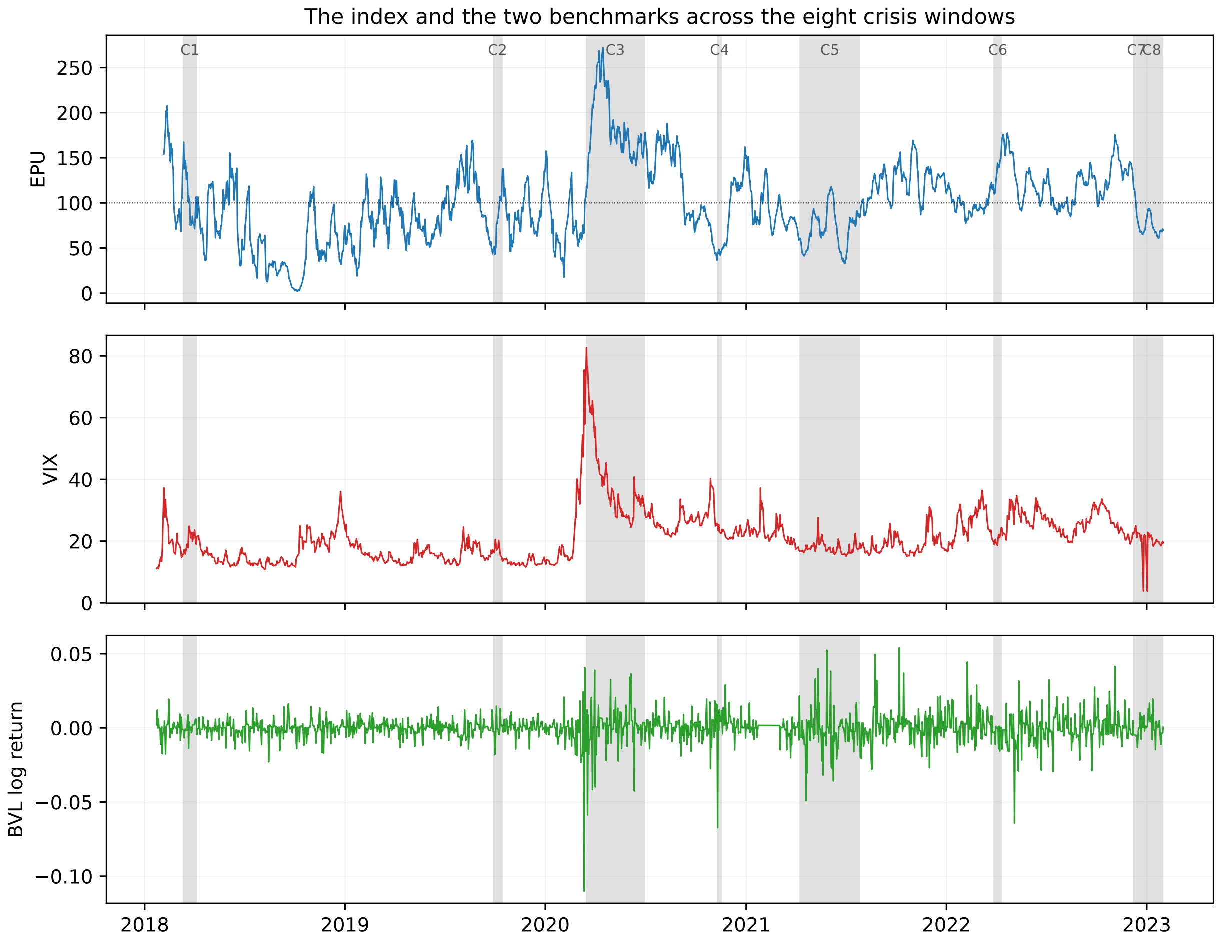

Figure 8. The Index and the Two Benchmarks Across the Eight Crisis Windows.

Top to bottom: the headline EPU, the VIX, and the Lima-exchange log return, with the eight domestic crisis windows C1–C8 (2018–2023) shaded. The index rises inside the shaded windows, alongside spikes in the VIX and in market-return volatility.

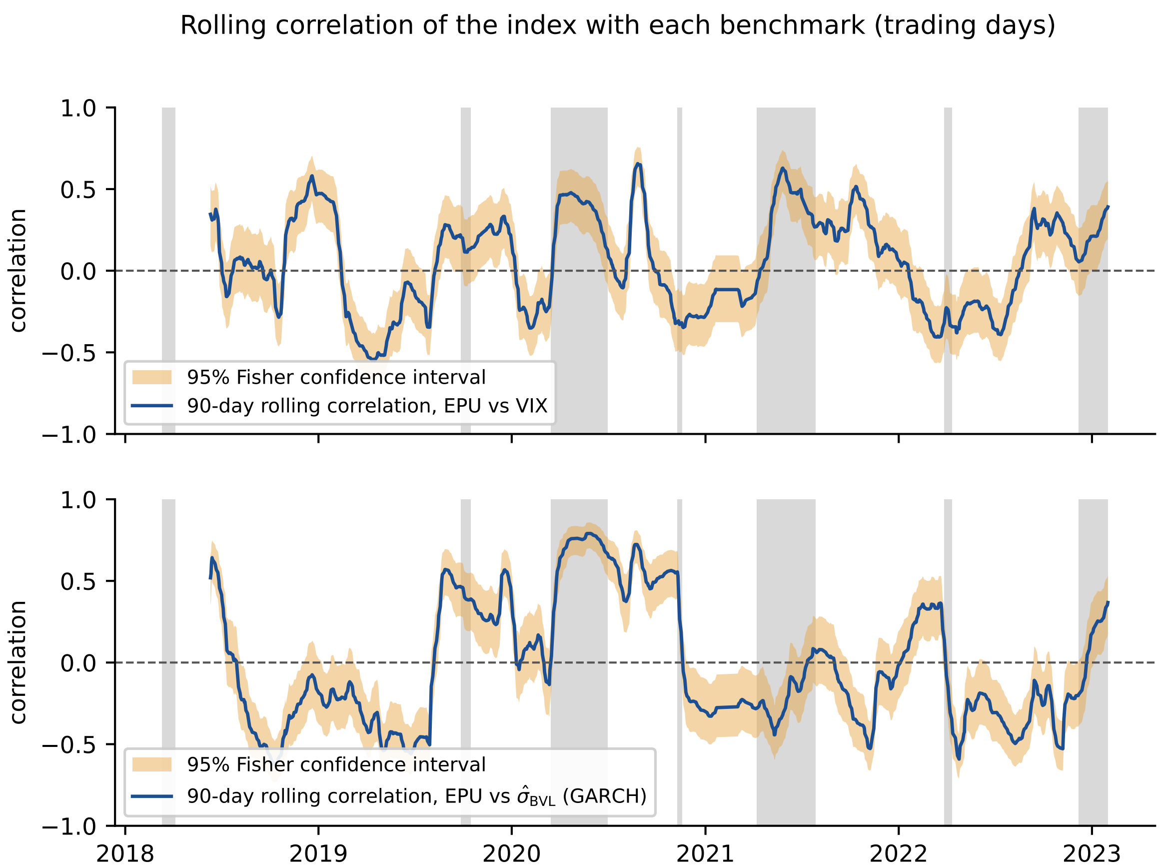

Figure 9. Rolling 90-Day Correlation of the EPU Index with the VIX and with the Filtered BVL Volatility.

Note: rolling 90-trading-day Pearson correlation of the headline index with the VIX (top) and with the GARCH conditional volatility of the Lima return (bottom), on the common trading-day panel. The legend names both objects explicitly and they are drawn in different colours: the dark solid line is the correlation itself, and the shaded band in the contrasting colour is its 95 percent interval from the Fisher transformation. The dashed horizontal rule marks zero and the grey vertical bands are the crisis windows of Table 7. The panel is computed on the trading-day panel, so it is comparable with Table 10; the previous version drew it on the interpolated calendar panel and against RV_t rather than the filtered volatility.

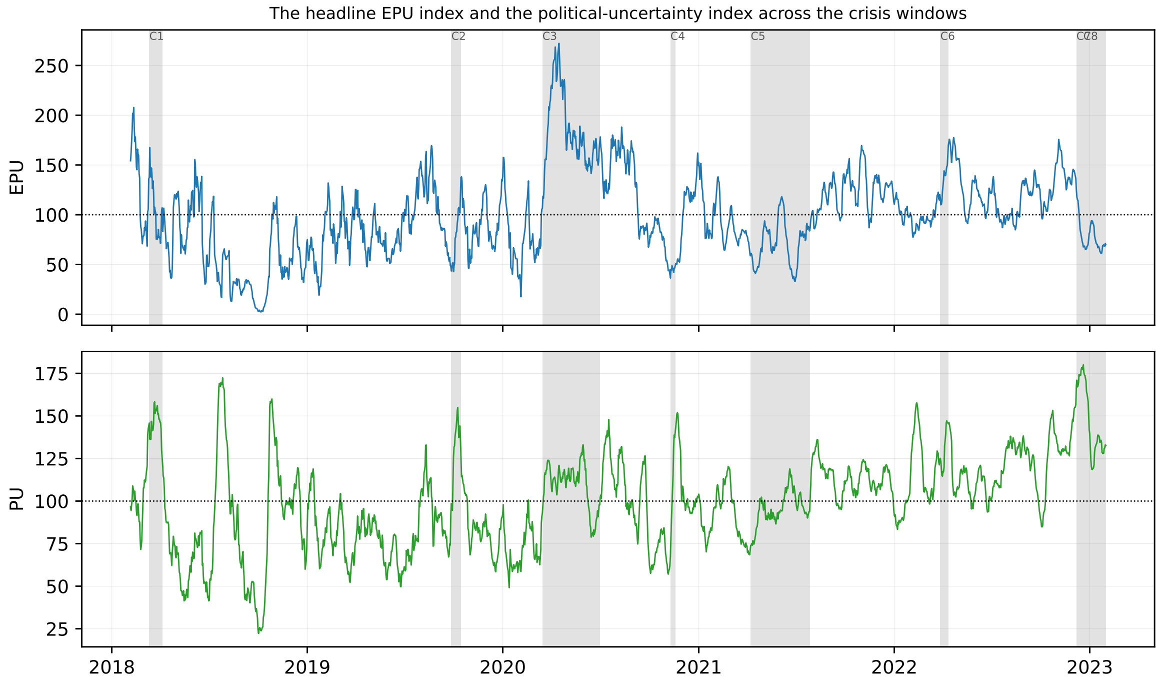

Figure 10. The Headline Index and the Political-Uncertainty Index Across the Crisis Windows.
Top: the headline EPU index of Eq. (5). Bottom: the political-uncertainty (PU) index, the same volume-weighted 14-day rate applied to the $P \cap U$ cell and rescaled to base 100. Shaded bands are the crisis windows C1–C8 of Table 7; the dotted line marks the base of 100. The headline index peaks at 272 on 15 April 2020, inside C3, and falls below its base in the purely political ruptures C4 and C7, where the PU index instead averages 174 in C7. The two indices resolve different crises: the strict cell requires an economic frame that an institutional collapse does not supply.

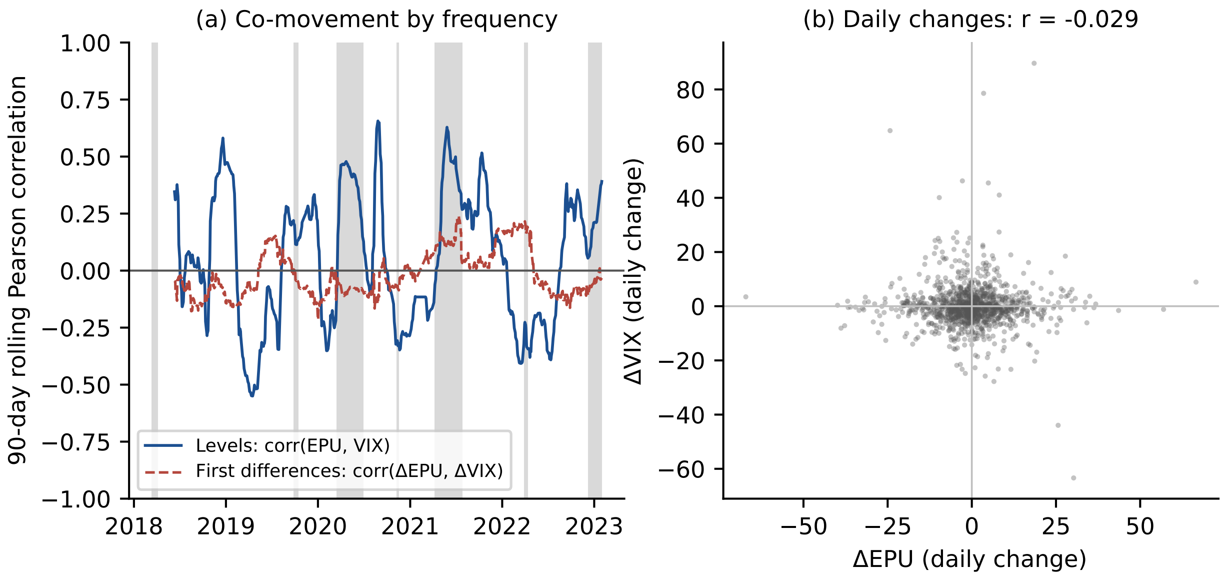

Figure 11. Level Co-Movement and First-Difference Co-Movement of the Index and the VIX.

Note: (a) 90-day rolling Pearson correlation of Eq. (10) between the index and the VIX in levels (solid) and in first differences (dashed), with the crisis windows C1–C8 shaded; (b) scatter of the daily change in the index against the daily change in the VIX. The level correlation swings with the crisis regimes while the difference correlation stays near zero throughout (−0.012 over the sample), so the similarity is a low-frequency, between-episode agreement rather than a daily one.

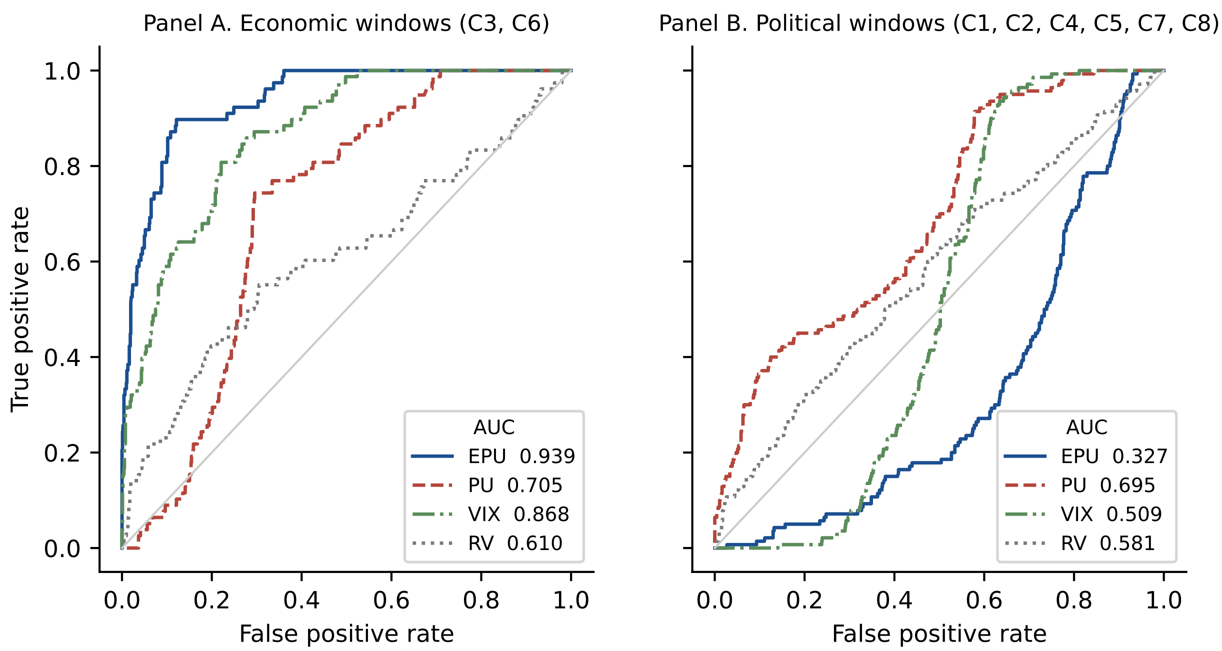

Figure 12. Receiver Operating Characteristic Curves for Crisis Days.

Note: Panel (a) pools the economic windows C3 and C6 of Table 7 (78 trading days); Panel (b) pools the six political windows (140 trading days). In both panels the negative class is the same 924 calm trading days, and the curves are computed on the trading-day panel of Section 4.1, so they reproduce Table 9 exactly. Each curve traces the true positive rate against the false positive rate as the classification threshold on the series moves across its support; the diagonal is the coin flip. Legend entries report $\widehat{\mathrm{AUC}}$. The headline index runs above the diagonal in Panel (a) and below it in Panel (b); the political-uncertainty index runs above the diagonal in both.

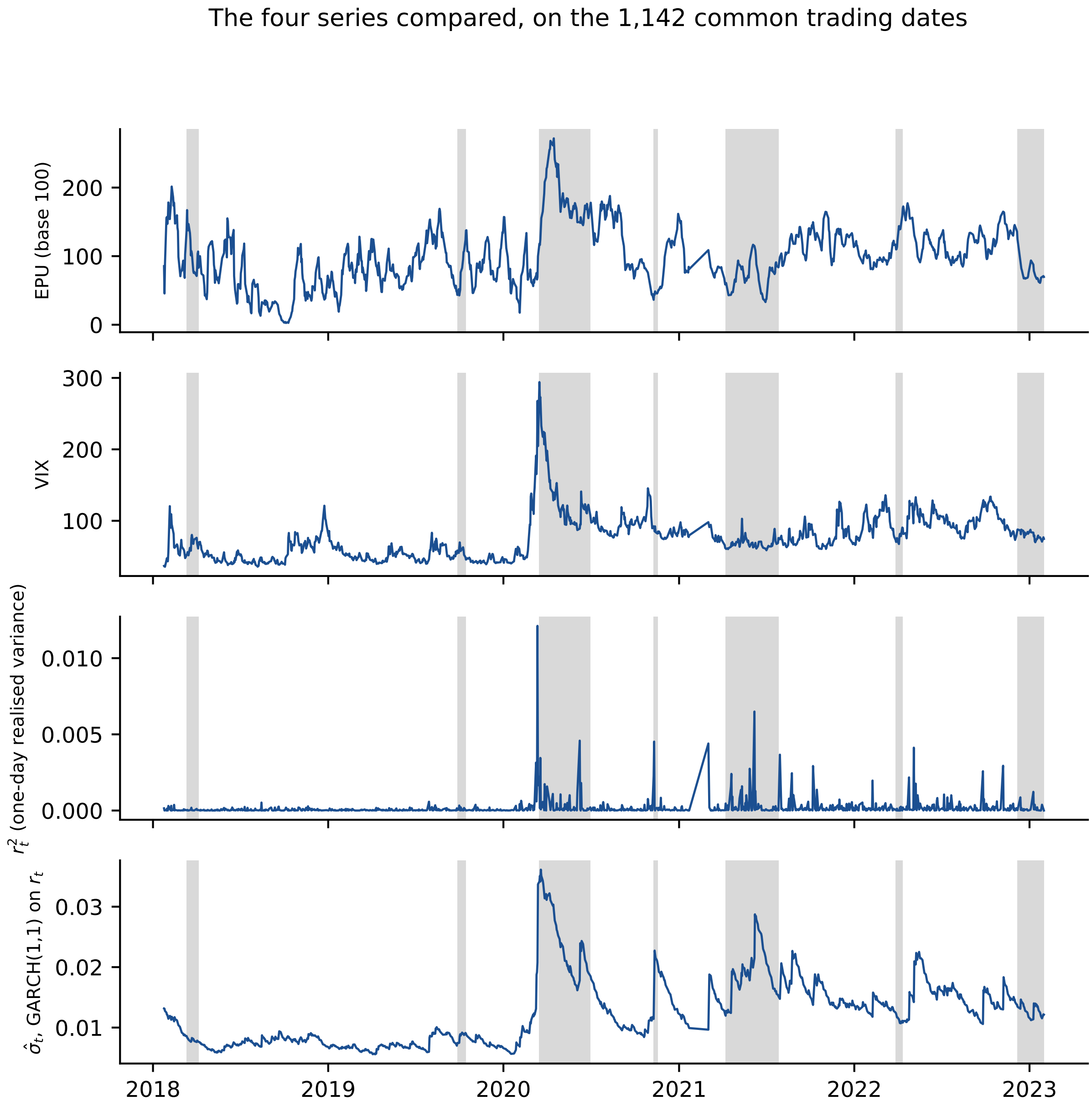

Figure 13. The Four Series Compared, on the Common Trading-Day Panel

Note: the headline index, the VIX, the one-day realised variance r_t^2, and the conditional volatility of the Lima return from the GARCH$(1,1)$ of Table 5, all on the 1 143 common trading dates. Grey bands are the crisis windows of Table 7. The contrast between the third and fourth panels is the point: r_t^2 is a spike process that reverts within a day, while the filtered volatility moves at the fortnightly horizon of the index.

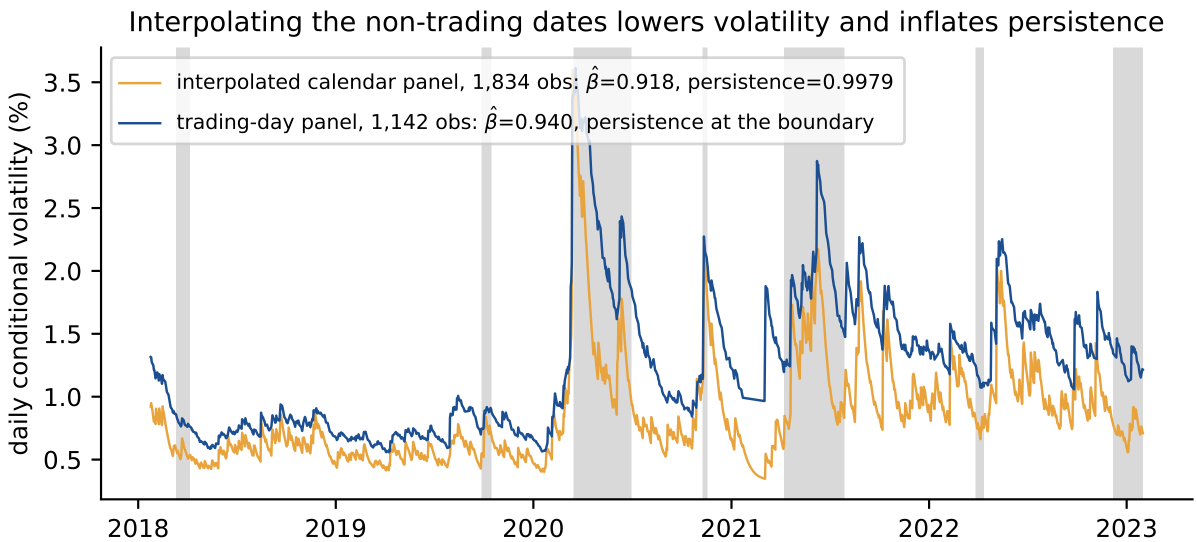

Figure 14. What Interpolating the Non-Trading Dates Does to the Volatility Filter

Note: conditional volatility of the BVL log-return from the same Gaussian GARCH(1, 1) estimated twice, once on the interpolated calendar panel of 1 834 dates and once on the 1 142 genuine trading dates. Interpolation lowers the level of the filtered volatility throughout and raises the estimated persistence, as Table 5 records.

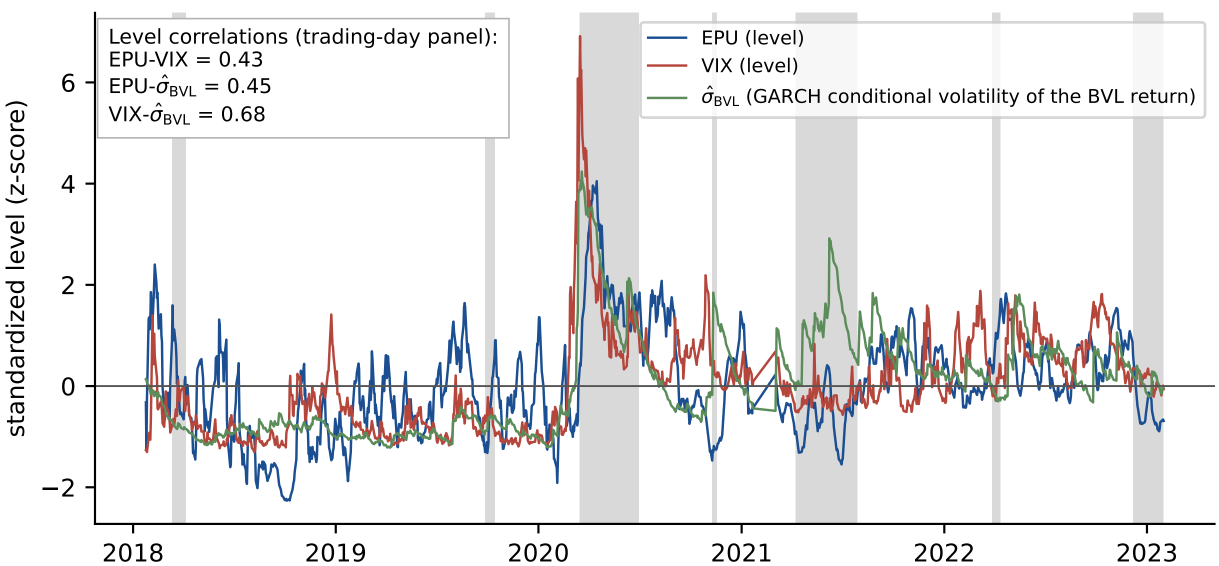

Figure 15. Three Levels of Uncertainty on the Same Plane.

Standardized levels of the headline index, the VIX, and the conditional volatility of the BVL return $\hat{\sigma}_{\mathrm{BVL},t}$ from the GARCH$(1,1)$ model of Table 5, over the $T = 1\,142$ trading-day panel of Section 4.1, with the crisis windows C1–C8 of Table 7 shaded. The GARCH filter is used only to turn the BVL return into a volatility level; the VIX enters as its own level and the index as its own level. The three series rise together in the crisis windows, most sharply in the pandemic window C3, and settle together outside them. The pairwise correlations are in Table 10.

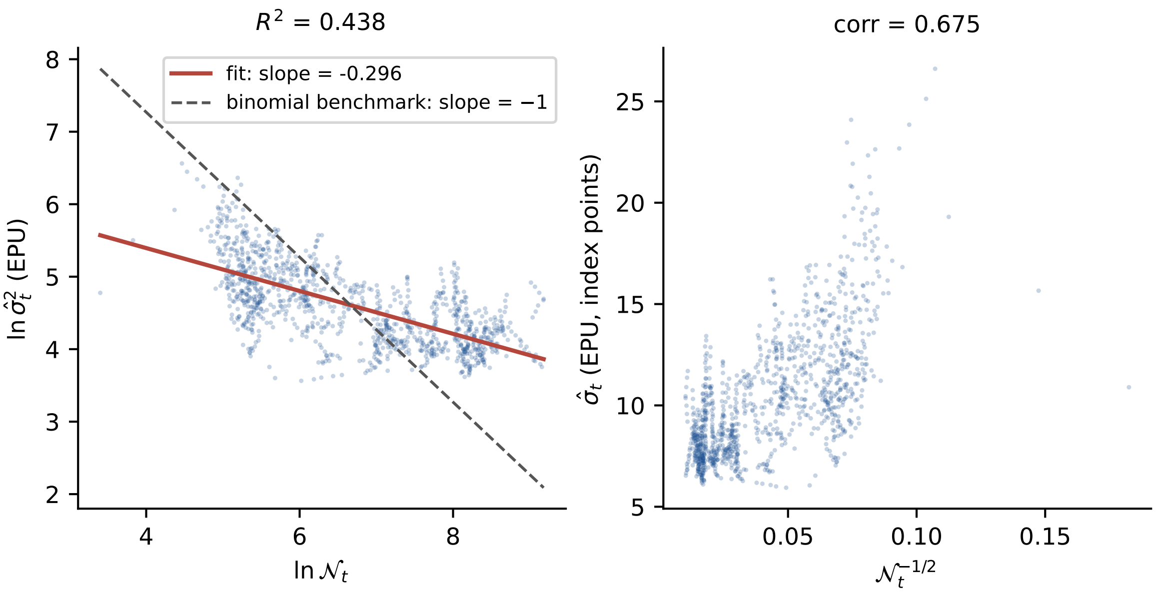

Figure 16. The Conditional Volatility of the Index and the Precision of Its Measurement.

Left: $\ln \hat{\sigma}_t^2$ of the index against $\ln \mathcal{N}_t$, the logarithm of the trailing fourteen-day tweet volume, with the fitted line; the slope is -0.296 and $R^2 = 0.438$, against the slope of -1 that a pure binomial sampling term would deliver. Right: $\hat{\sigma}_t$ against $\mathcal{N}_t^{-1/2}$, the shape of the sampling standard error of the rate ρ_t of Eq. (3); the correlation is 0.675. Both panels are computed on the trading-day panel of Section 4.1. Roughly two fifths of the movement in the conditional variance of the index is the precision with which the day's rate is estimated.

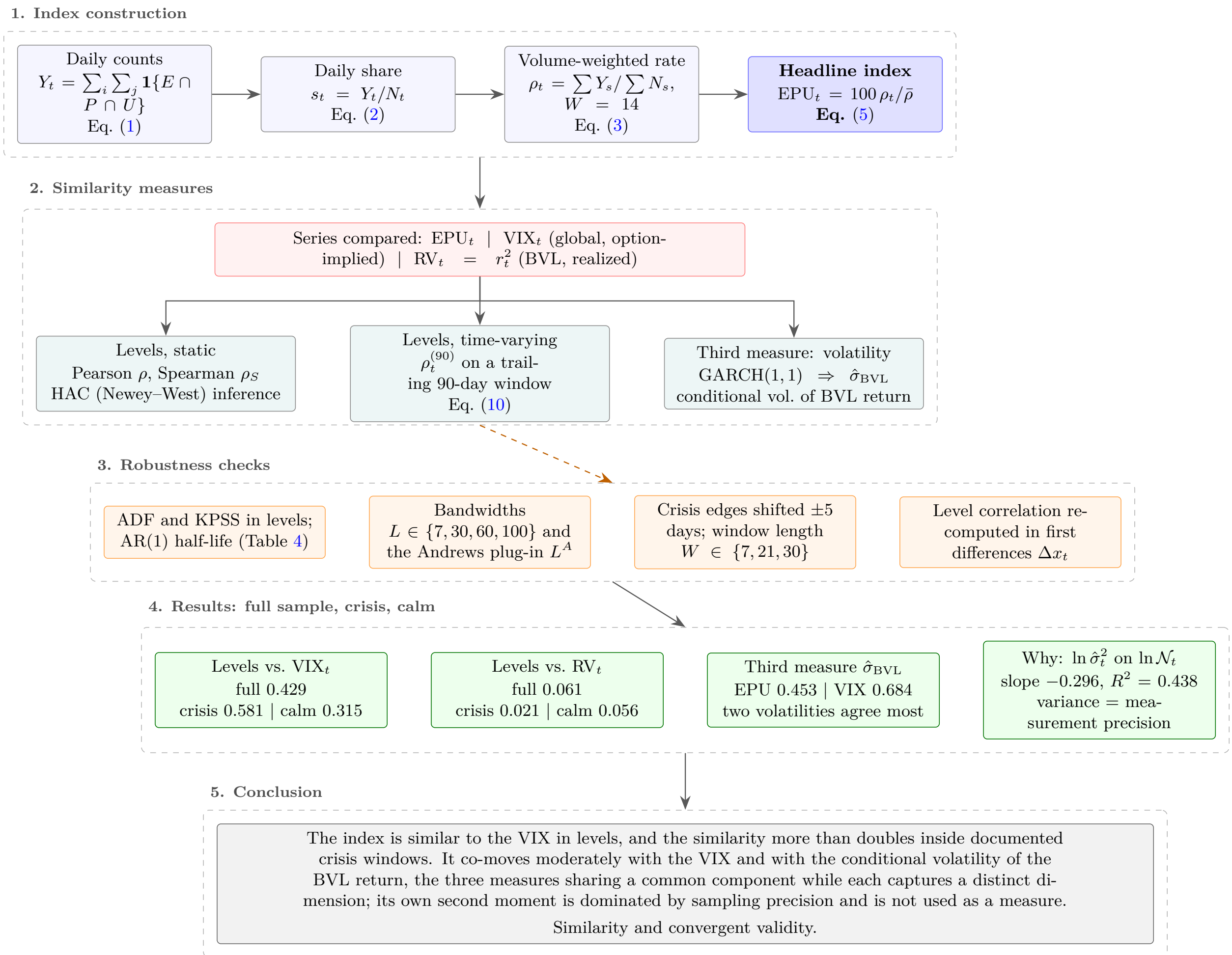

Figure 17. Synthesis of the Paper.

The construction of the headline index (top), the three measures of similarity applied to the two benchmarks (middle left), the diagnostics that discipline them (middle right), the results on the full sample and on the crisis and calm subsamples (bottom), and what the evidence licenses (foot). Solid arrows carry data; dashed arrows carry robustness checks. Except where noted (the 14-day window, the HAC bandwidths, and the volume elasticity -0.296), the reported numbers are Pearson correlations computed on the frozen daily series of Section 2.